ACC501 – Business Finance
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Highlighted by MR Shahzad

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LESSON 1

BUSINESS FINANCE

The Primary textbook for the course is:

- **Essentials of Corporate Finance,** by Ross, Westerfield and Jordan, fourth edition, McGraw Hill Publishers, ISBN 0-07-121057-7

Reference books will be:

- **Introduction to Finance** by Lawrence J. Gitman and Jeff Madura, Addison-Wesley Publishers
- **Foundations of Financial Management** by Stanley B. Block and Geoffrey A. Hirt, McGraw Hill Publishers

Course Contents

- An overview of Financial Environment
- Financial Statements, Taxes and Cash Flows
- Time Value of Money and Discounted Cash Flow Valuation
- Valuation of Stocks and Bonds
- Net Present Value and other Investment Criteria
- Capital Investment Decision
- Risk and Return
- Cost of Capital
- Leverage and Capital Structure
- Raising Capital
- Working Capital Management
- Dividends

Highlighted by MR Shahbaz

Finance: A Quick Look

- Four Basic Areas
  - Business Finance
  - Investments
  - Financial Institutions
  - International Finance

Business Finance

Addresses the following three questions:

- What long-term investments should the firm engage in?
- How can the firm raise the money for the required investments?
- How much short-term cash flow does a company need to pay its bills?

Investments

- Deals with financial assets such as stocks and bonds.
- It covers the following issues:
  - Pricing Financial Assets
  - Associated Risks and Rewards
  - Determining best mixture of financial investment
- Career opportunities in investment
  - Stock Brokerage

- Portfolio Management
- Security Analysis

Financial Institutions

- Businesses dealing in financial matters
  - Banks and Insurance companies

International Finance

- Covers international aspects of corporate finance, investment and financial institutions.

Highlighted by MR Shahbaz

LESSON 2

WHY STUDY FINANCE?

Why Study Finance?

- Marketing and Finance
  - Marketers have to work with budgets
  - Need to get greatest payoffs from marketing expenditures and programs
  - Cost and Benefit analysis of projects
  - So, finance is vital for:
    - Marketing research
    - Design of marketing and distributions channels
    - Product pricing
- Accounting and Finance
  - Accountants are required to make financial decisions as well as understand the implications of new financial contracts
  - Financial analysts make extensive use of accounting information
- Management and Finance
  - Business Strategy is always disastrous if financial planning is not adhered to.

What is Business Finance?

- In order to start any new business, the following issues become vital
  - What long-term investment should be taken on?
  - From where to get the long-term financing to pay for investment? Bring in other owners or borrow the money?
  - How to manage everyday financial activities?

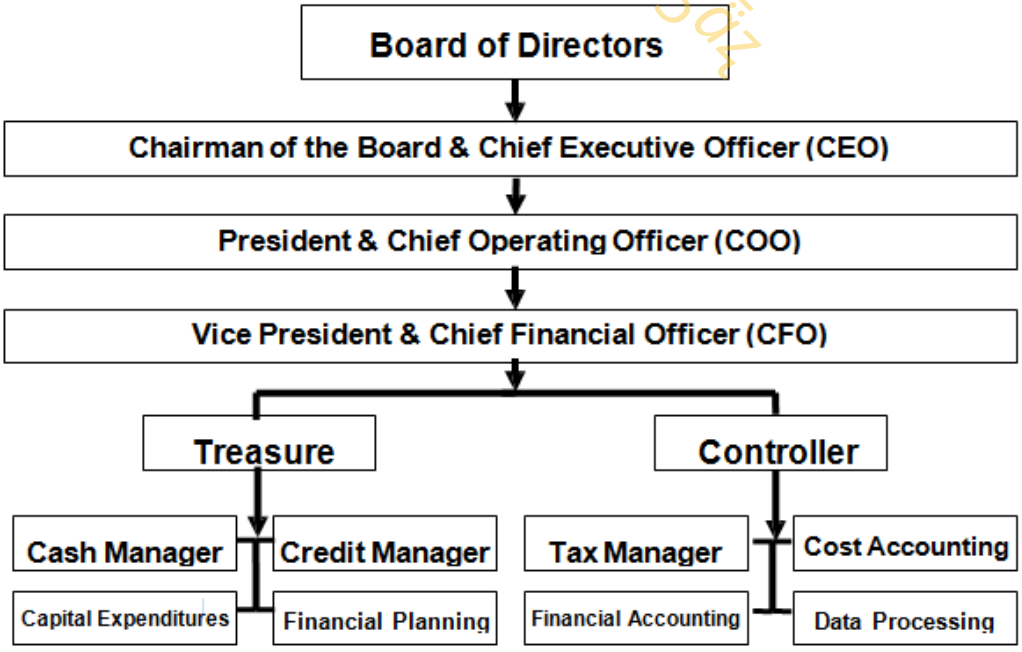
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The Financial Manager

To create value, the financial manager should:

- Try to make smart investment decisions.
- Try to make smart financing decisions.

Hypothetical Organization Chart

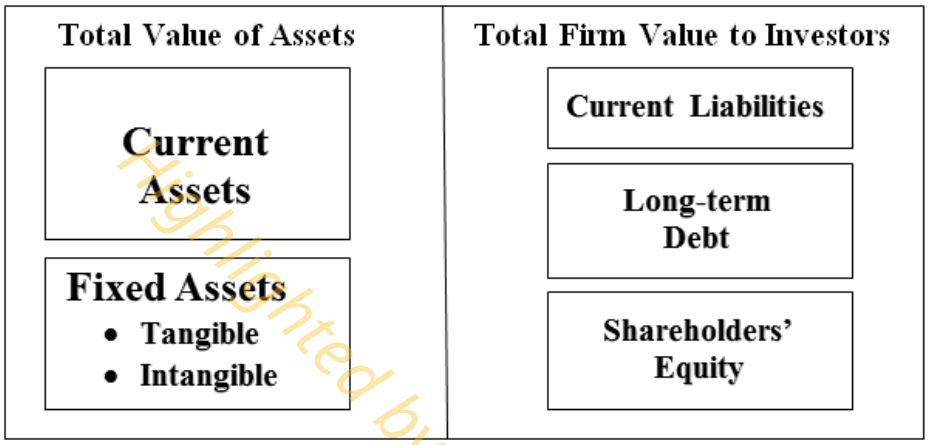


Business Finance and Financial Manager

- Financial Management Decisions
  - Capital Budgeting
  - Capital Structure
  - Working Capital Management

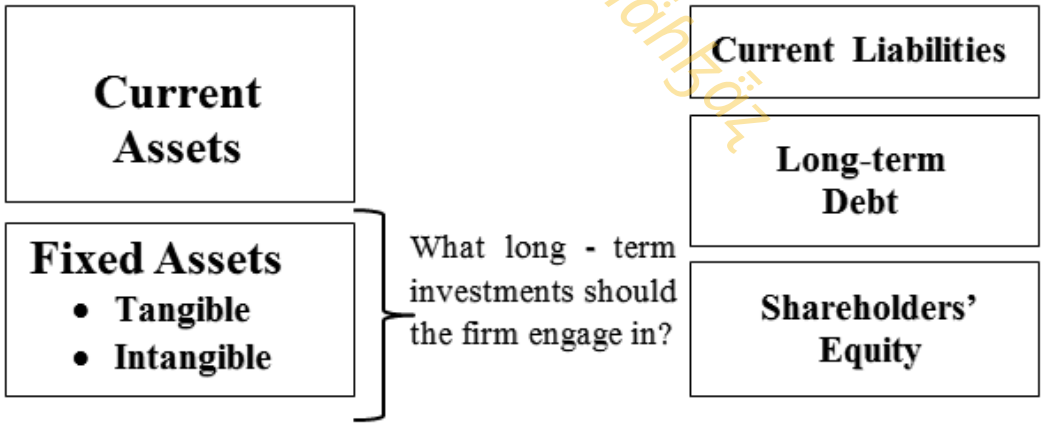
Financial Management Decisions

- Capital Budgeting
- The process of planning and managing a firm's long-term investments
- Financial managers concern with how much, when and how likely is cash expected to receive
- Evaluating the size, timing and risk of future cash flows is the essence of capital budgeting

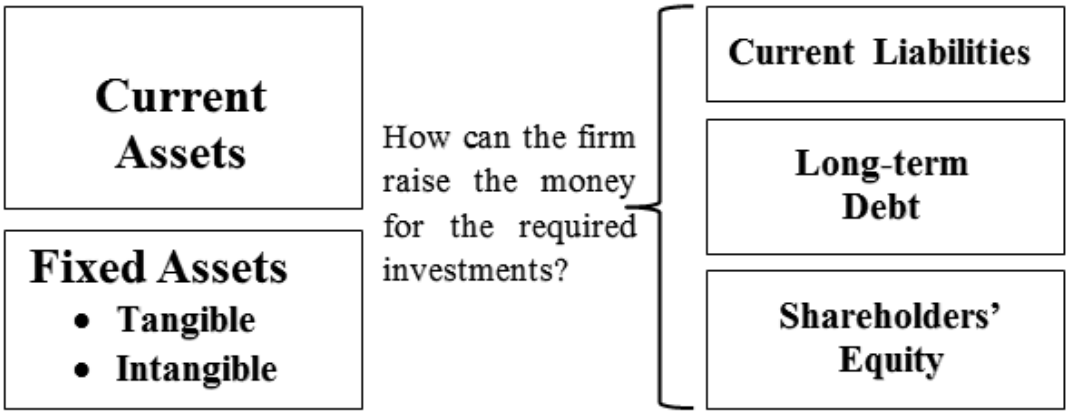


Highlighted by MR Shahbaz

Capital Budgeting Decision

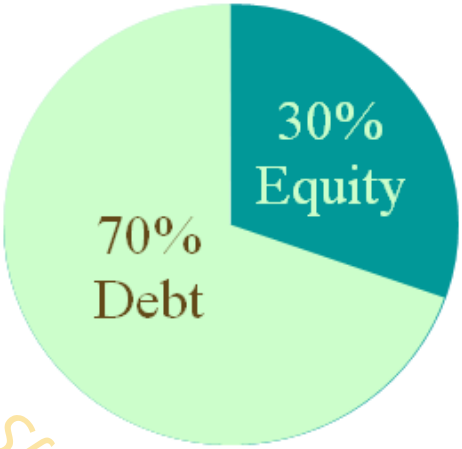


The Capital Structure Decision



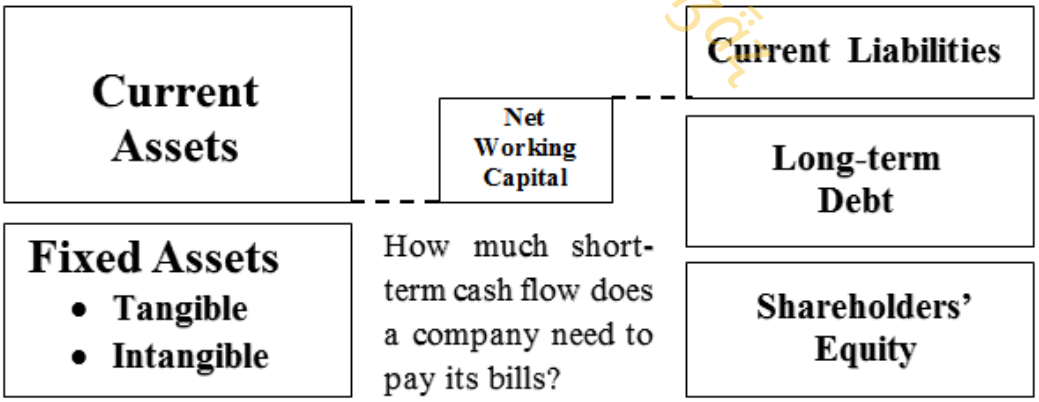
Capital Structure

- The value of the firm can be thought of as a pie.
- The goal of the manager is to increase the size of the pie.
- The Capital Structure decision can be viewed as how best to slice up the pie.
- If how you slice the pie affects the size of the pie, then the capital structure decision matters.



Highlighted by MR Shahbaz

The Net Working Capital Investment Decision



The Corporate Firm

- The corporate form of business is the standard method for solving the problems encountered in raising large amounts of cash.
- However, businesses can take other forms.

Forms of Business Organization

- Three major forms
  - Sole proprietorship
  - Partnership
    - General
    - Limited
  - Corporation
    - Limited Liability Company

Sole Proprietorship

- Advantages
  - Easiest to start
  - Least regulated
  - Single owner keeps all the profits
  - Taxed once as personal income
- Disadvantages
  - Limited to life of owner
  - Equity capital limited to owner's personal wealth
  - Unlimited liability
  - Difficult to sell ownership interest

Partnership

- Two or more owners (partners)
  - General partnership: all partners share in gains and losses and all have unlimited liability for all partnership debts
  - Limited partnership: one or more general partners will run the business and have unlimited liability but there will be one or more limited partners who do not actively participate in the business and their liability is limited to their contribution.
- Advantages
  - Two or more owners
  - More capital available
  - Relatively easy to start
  - Income taxed once as personal income
- Disadvantages
  - Unlimited liability
    - General partnership
    - Limited partnership
  - Partnership dissolves when one partner dies or wishes to sell
  - Difficult to transfer ownership

Highlighted by MR Shahbaz

LESSON 3

THE CORPORATE FIRM

The corporate form of business is the standard method for solving the problems encountered in raising large amounts of cash. However, businesses can take other forms.

Forms of Business Organization

- Three major forms
  - Sole proprietorship
  - Partnership
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Highlighted by MR Shafiq

Partnership

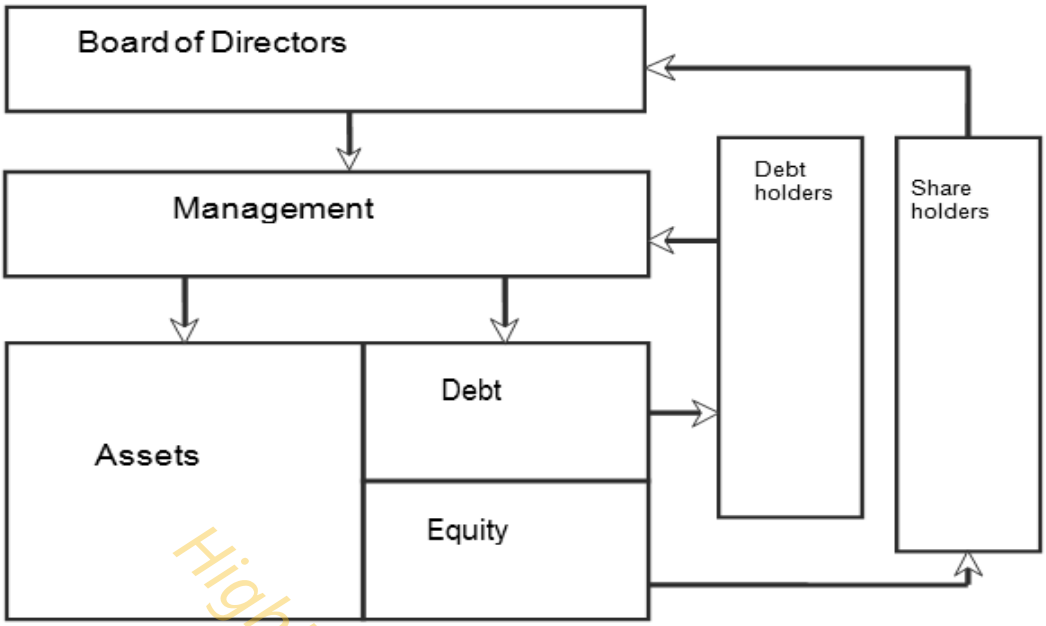
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- Disadvantages
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    - General partnership
    - Limited partnership
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  - Difficult to transfer ownership

Corporation

- A business created as a distinct legal entity owned by one or more individuals or entities.
- Forming of corporation involves preparing
  - Charter including corporation's name, intended life, business purpose and number of shares

- Set of bylaws which describes the regulations for the business

Separation of Ownership and Control



Corporation

- Advantages
  - Limited liability
  - Unlimited life
  - Separation of ownership and management
  - Transfer of ownership is easy
  - Easier to raise capital
- Disadvantages
  - Separation of ownership and management
  - Double taxation (income taxed at the corporate rate and then dividends taxed at personal rate)

Goal of the Corporate Firm

- The traditional answer is that the managers of the corporation are obliged to make efforts to maximize shareholder wealth.
- Alternatively, the goal of the financial manager is to maximize the current value per share of the existing stock.

The Set-of-Contracts Perspective

- The firm can be viewed as a set of contracts.
- One of these contracts is between shareholders and managers.
- The managers will usually act in the shareholders' interests.
  - The shareholders can devise contracts that align the incentives of the managers with the goals of the shareholders.
  - The shareholders can monitor the managers' behavior.
- This contracting and monitoring is costly.

Highlighted by MR Shahzad

The Agency Problem

- Agency relationship
  - Principal hires an agent to represent their interest
  - Stockholders (principals) hire managers (agents) to run the company
- Agency problem
  - Conflict of interest between principal and agent
- Management goals and agency costs

Managerial Goals

- Managerial goals may be different from shareholder goals
  - Expensive perquisites
  - Survival
  - Independence
- Increased growth and size are not necessarily the same thing as increased shareholder wealth.

Do Shareholders Control Managerial Behavior?

- Shareholders vote for the board of directors, who in turn hire the management team.
- Contracts can be carefully constructed to be incentive compatible.
- There is a market for managerial talent—this may provide market discipline to the managers—they can be replaced.
- If the managers fail to maximize share price, they may be replaced in a hostile takeover.

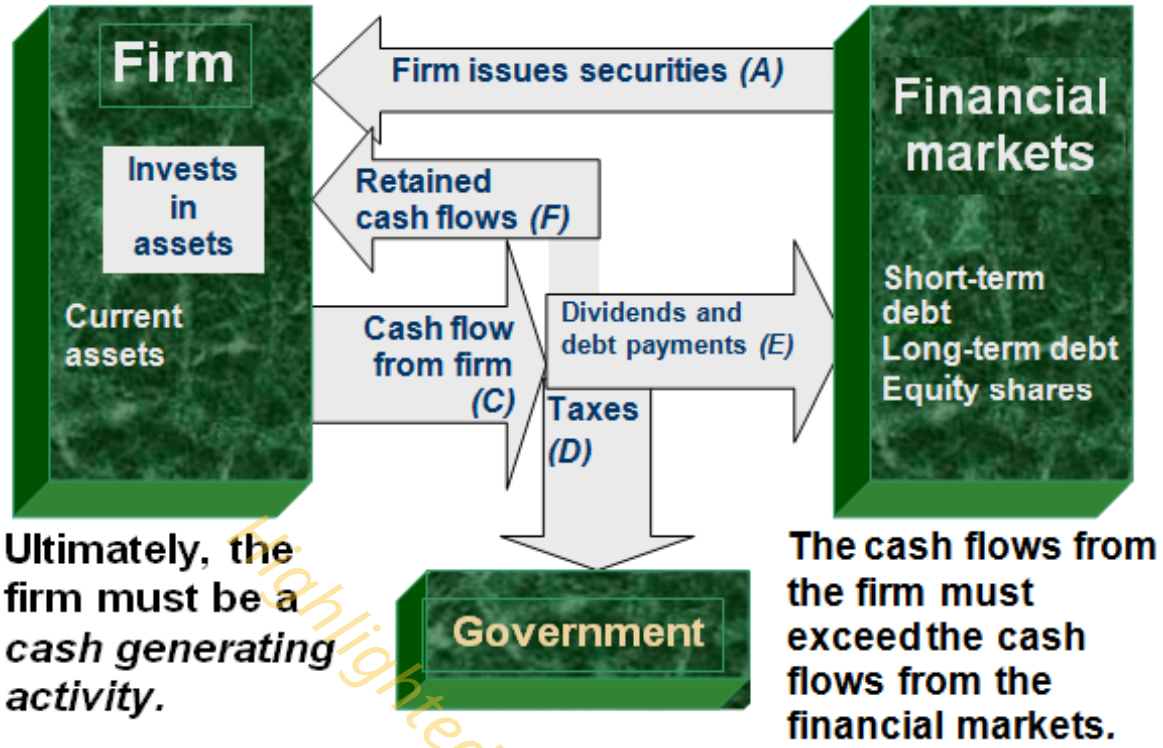
Managing Managers

- Managerial compensation
  - Incentives can be used to align management and stockholder interests
  - The incentives need to be structured carefully to make sure that they achieve their goal
- Corporate control
  - The threat of a takeover may result in better management
- Other stakeholders

Highlighted by MR Shahzad

LESSON 4

THE FIRM AND THE FINANCIAL MARKETS

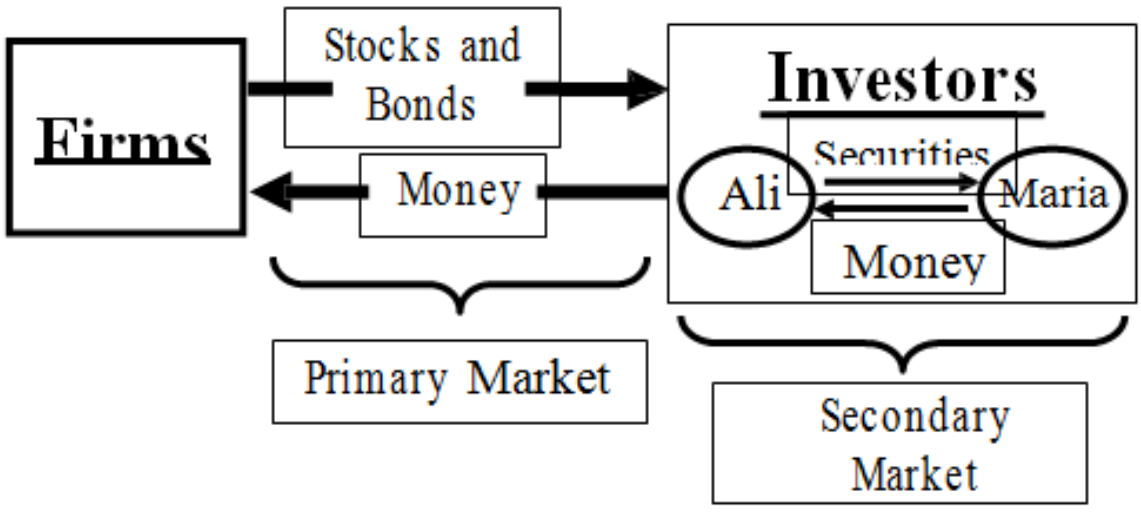


Highlighted by MB Shāfīq

Financial Markets

- Primary Market
  - When a corporation issues securities, cash flows from investors to the firm.
  - Usually an underwriter is involved
- Secondary Markets
  - Involve the sale of "used" securities from one investor to another.
  - Securities may be exchange traded or trade over-the-counter in a dealer market.

Financial Markets



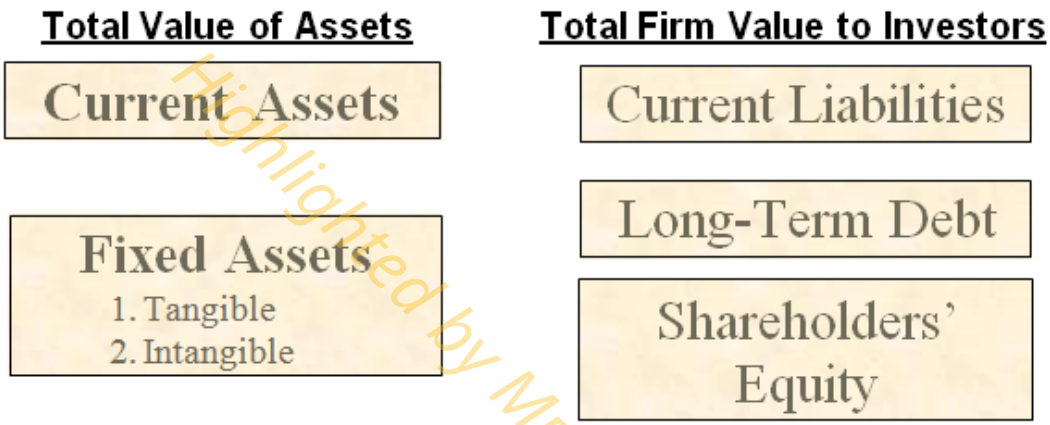
Dealer vs. Auction Markets

- Auction markets are different from dealer markets in two ways:
  - Trading in a given auction exchange takes place at a single site on the floor of the exchange.
  - Transaction prices of shares are communicated almost immediately to the public.
  - Listing

The Balance Sheet

- An accountant's snapshot of the firm's accounting value as of a particular date.
- The Balance Sheet Identity is:
  Assets ≡ Liabilities + Stockholder's Equity
- When analyzing a balance sheet, the financial manager should be aware of three concerns:
  Accounting Liquidity, Debt versus Equity, and Value versus Cost

The Balance-Sheet Model of the Firm



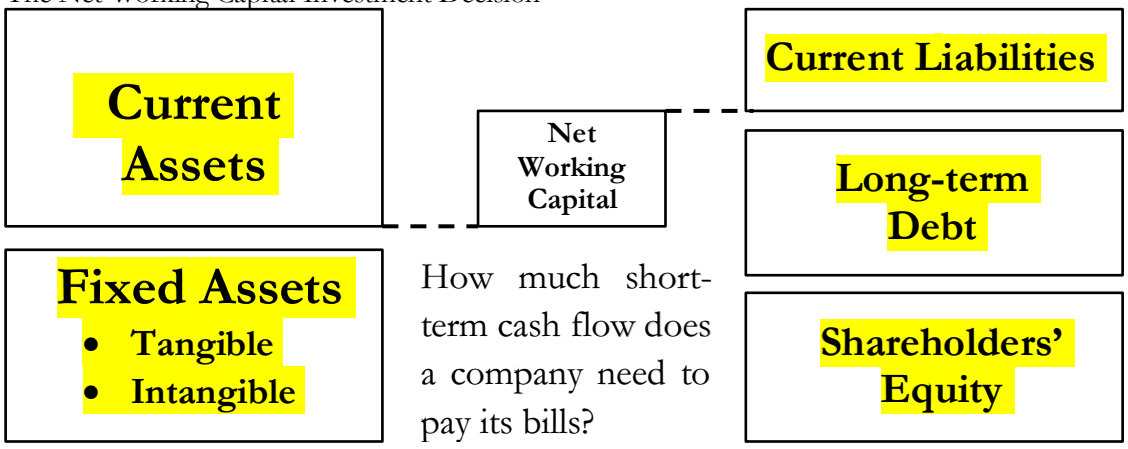
Highlighted by MR Shahzad

Net Working Capital

- Net Working Capital = Current Assets – Current Liabilities
  - NWC > 0 when Current Assets > Current Liabilities
  - NWC < 0 when Current Assets < Current Liabilities
  - NWC = 0 when Current Assets = Current Liabilities
- NWC usually grows with the firm for the healthy firms.

The Balance-Sheet Model of the Firm

The Net Working Capital Investment Decision



Building the Balance Sheet

- A firm has:
  - Current Assets of $100,
  - Net Fixed Assets of $500,
  - Short-term Debt of $70, and
  - Long-term Debt of $200
- Now…
  - Total Assets are $100 + 500 = $600
  - Total Liabilities are $70 + 200 = $270
  - Shareholders' Equity is $600 – 270 = $330

Building the Balance Sheet

Assets		Liabilities and Shareholders' Equity	
Current Assets	$100	Current Liabilities	$70
Net Fixed Assets	500	Long Term Debt	200
		Shareholders' equity	330
Total Assets	**$600**	**Total liabilities and Shareholders' equity**	**$600**

The Balance Sheet of the XYZ Corporation

XYZ CORPORATION
Balance Sheet
20X2 and 20X1
(in $ millions)

Assets	20X2	20X1	Liabilities (Debt) and Stockholder's Equity	20X2	20X1
Current assets			**Current Liabilities**		
Cash and equivalents	$140	$107	Accounts payable	$213	$197
Accounts receivable	294	270	Notes payable	50	53
Inventories	269	280	Accrued expenses	223	205
Other	58	50			
Total current assets	$761	$707	Total current liabilities	$486	$455
Fixed assets			**Long-term liabilities**		
Property, plant, and equipment	$1423	$1274	Deferred taxes	$117	$104
Less accumulated depreciation	-550	-460	Long-term debt	471	458
Net property, plant, and equipment	$873	$814	Total long-term liabilities	$588	$562
Intangible assets and other	245	221	**Stockholder's equity**		
Total fixed assets	$1,118	$1,035	Preferred stock	$39	$39
			Common stock ($1 per value)	55	32
			Capital surplus	347	327
			Accumulated retained earnings	390	347
			Less treasury stock	-26	-20
			Total equity	$805	$725
Total assets	**$1,879**	**$1,742**	**Total liabilities and stockholder's equity**	**$1,879**	**$1,742**

Highlighted by MR Shahzad

Notes:

1. Long-term debt rose by ($471 million–$458 million) $13 million. This is the difference between $86 million new debt and $73 million in retirement of old debt.

2. Treasury stock rose by $6 million. This reflects the repurchase of $6 million stock.

3. XYZ Corporation reports $43 million in new equity. The company issued 23 million shares at a price of $1.87. The par value of common stock is increased by $23 million and capital surplus is increased by $20 million.

Balance Sheet Analysis

- When analyzing a balance sheet, the financial manager should be aware of three concerns:
  - Accounting liquidity
  - Debt versus equity
  - Value versus cost

Accounting Liquidity

- Refers to the ease and quickness with which assets can be converted to cash.
- Current assets are the most liquid.
- Some fixed assets are intangible.
- The more liquid a firm's assets, the less likely the firm is to experience problems meeting short-term obligations.
- Liquid assets frequently have lower rates of return than fixed assets.

The Balance Sheet of the XYZ Corporation

XYZ CORPORATION
Balance Sheet
20X2 and 20X1
(in $ millions)

$252m = $707- $455

Assets	**20X2**	**20X1**	**Liabilities (Debt) and Stockholder's Equity**	**20X2**	**20X1**
Current assets			**Current Liabilities**		
Cash and equivalents	$140	$107	Accounts payable	$213	$197
Accounts receivable	294	270	Notes payable	50	53
Inventories	269	280	Accrued expenses	223	205
Other	58	50			
Total current assets	**$761**	**$707**	**Total current liabilities**	**$486**	**$455**

Here we see NWC grow to $275 million in 20X2 from $252 million in 20X1

$23

$275m = $761m - $486

This increase of $23 million is an investment of the firm.

Debt versus Equity

- Generally, when a firm borrows it gives the bondholders first claim on the firm's cash flow.
- Thus shareholder's equity is the residual difference between assets and liabilities.

  Shareholders' Equity = Assets – Liabilities
- The Use of debt in a firm's capital structure is called "Financial Leverage"
  - The more debt a firm has (as a percentage of assets) the greater is the degree of financial leverage
  - Debt acts as a lever in the sense that it magnifies both gains and losses

Highlighted by Muhtaz

Value versus Cost

- The true value of any asset is its market value, which is simply the amount of cash we would get if we actually sold it.
- The values shown on the balance sheet for the firm's assets are book values and generally are not what the assets are actually worth.
- Under the Accounting standards audited financial statements of firms carry assets at historical cost.
- For current assets, market value and book value might be somewhat similar since they are bought and converted into cash over a relatively short span of time.
- For fixed assets, it's very unlikely that the actual market value of an asset is equal to its book value.
  - Example: Land purchased for railroads a century ago
- Similarly the owner's equity figure on the balance sheet and the true market value of the equity need not be related.
- For Financial Managers, accounting value of the equity is not a matter of concern rather it is the market value of the shares that matters.

Market vs. Book Value

- K Corporation has fixed assets with a book value of $700 and an appraised market value of $1,000
- Net working capital is $400 on the books but approx. $600 would be realized if the current accounts were liquidated
- K has $500 in long-term debt, both book & market value
  - What is the book value of the equity?
  - What is the market value?

Highlighted by Mr. Shoaib

K Corporation
Balance Sheet
Market Value vs. Book Value

Assets	Book	Market	Liabilities (Debt) and Stockholder's Equity	Book	Market
Net working Capital	$400	$600	Long-term debt	$500	$500
Net Fixed Assets	$700	$1,000	Shareholders' equity	$600	$1,100
TOTAL	$1,100	$1,600	**TOTAL**	$1,100	$1,600

LESSON 5

THE INCOME STATEMENT

The Income Statement

- If we think of the balance sheet as a snapshot then we can think of income statement as a video recording covering before and after the picture.
- The income statement measures performance over a specific period of time.
- The accounting definition of income is:

Revenue – Expenses ≡ Income

XYZ Corporation Income Statement

The operations section of the income statement reports the firm's revenues and expenses from principal operations.

XYZ CORPORATION Income Statement 20X2 (in $ millions)	
Total operating revenues	$2,262
Cost of goods sold	- 1,655
Selling, general, and administrative expenses	- 327
Depreciation	-90
Operating income	$190
Other income	29
Earnings before interest and taxes	$219
Interest expense	-49
Pretax income	$170
Taxes	-84
Current: $71	
Deferred: $13	
Net income	$86
Retained earnings	$43
Dividends	$43

Highlighted by MR Shahbaz

The non- operating section of the income statement includes all financing costs, such as interest expense.

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Usually a separate section reports as a separate item the amount of taxes levied on income.

Highlighted by MR Shahbaz

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Net income is the "bottom line".

Income Statement Analysis

- There are three things to keep in mind when analyzing an income statement:
  - Generally Accepted Accounting Principles (GAAP)
  - Non-Cash Items
  - Time and Costs

Income Statement Analysis

- Generally Accepted Accounting Principles (GAAP)
  - "The Realization principle" is to recognize revenue when the earning process is complete, i.e. revenue is recognized at the time of sale, which need not be the same as time of collection.

- "The matching principal" of GAAP dictates that revenues be matched with expenses. Thus, income is reported when it is earned, even though no cash flow may have occurred.

Income Statement Analysis

- Non-Cash Items
  - The primary reason that accounting income differs from cash flow is that income statement contain non-cash items
  - Depreciation is the most apparent. No firm ever writes a check for "depreciation".
  - The depreciation deduction is simply an application of the matching principle in accounting.
  - Another noncash item is deferred taxes, which does not represent a cash flow.

Income Statement Analysis

- Time and Costs
  - In the short run, certain equipment, resources, and commitments of the firm are fixed, but the firm can vary such inputs as labor and raw materials.
  - In the long run, all inputs of production (and hence costs) are variable.
  - Financial accountants do not distinguish between variable costs and fixed costs. Instead, accounting costs usually fit into a classification that distinguishes product costs from period costs.

Highlighted by MR Shahbaz

Income Statement Analysis

- Time and Costs
  - Product cost include such things as raw materials, direct labor and manufacturing overhead and are reported on the income statement as the cost of goods sold, but they include both fixed and variable costs.
  - Period costs include selling, general, and administrative expenses which may be fixed as well as variable.

Taxes

- One of the largest cash outflows that a Corporate firm experiences.
- The size of tax is determined through the tax schedule issued by the Central Board of Revenue.
- Taxes for partnerships and proprietorship are computed using the personal income tax schedules.

Model Tax Rates

Taxable Income	Tax Rate
$ 0 – 50,000	15%
50,001 – 75,000	25
75,001 – 100,000	34
100,001 – 335,000	39
335,001 – 10,000,000	34
10,000,001 – 15,000,000	35
15,000,001 – 18,333,333	38
18,333,334 +	35

Average vs. Marginal Tax Rates

- Average tax rate is tax bill divided by the taxable income or the percentage of the income that goes to pay taxes
- Marginal tax rate is the extra tax you would pay if you earn one more dollar.
- Suppose a Corporation has a taxable income of $200,000. So the Tax calculation will be:

$50,000 × 15%	$7,500
($75,000 – 50,000) × 25%	6,250
($100,000 – 75,000) × 34%	8,500
($200,000 – 100,000) × 39%	39,000
TOTAL	**$ 61,250**

- Our total tax is $61,250
- Average tax rate is $61,250 / 200,000 = 30.625%
- Marginal rate is 39%

Flat Tax rate

- There is only one tax rate and this rate is same for all income levels.
- With such a tax, the marginal tax rate is always same as the average tax rate.
- The model tax rate schedule presented earlier represents a modified flat-rate tax, which becomes a true flat rate for the highest incomes.
- Let's take another view

Highlighted by Mr Shahzad

Average vs. Marginal Tax Rates

Taxable Income	Marginal Tax Rate	Total Tax	Average tax Rate
$ 45,000	15%	$ 6,750	15.00%
70,000	25	12,500	17.86
95,000	34	20,550	21.63
250,000	39	80,750	32.30
1,000,000	34	340,000	34.00
17,500,000	38	6,100,000	34.86
50,000,000	35	17,500,000	35.00
100,000,000	35	35,000,000	35.00

Average vs. Marginal Tax Rates

- We see that the more a corporation makes, the grater is the percentage of taxable income paid in taxes.
- So the average tax rate never goes down, even though the marginal rate does.
- It will normally be the marginal tax rate that is relevant for financial decision making, since any new cash flows will be taxed at the marginal rate

Cost of a Tax Deductible Expense

- The businesspersons often say that a tax-deductible item, such as interest on loans, travel expenditures, or salaries, costs substantially less than the amount spent on after-tax basis.
- Let's examine two corporations - one pays $100,000 in interest, while other has no interest expense.

	Corporation A	Corporation B
Earnings before interest and taxes	$400,000	$400,000
- Interest Expense	**100,000**	**0**
Earnings before taxes (taxable income)	300,000	400,000
- Taxes @35%	105,000	140,000
Earnings after taxes	$195,000	$260,000
Difference in earnings after taxes	**$65,000**	

It can also be computed as: Interest Expense (1 – Tax rate)
= $100,000 (1 – 35%)
= $65,000

Highlighted by MR Shahbaz

Cost of a Tax Deductible Expense

- Interest is deducted from earnings before determining taxable income, thus saving $35,000 in taxes and costing only $65,000 on a net basis.
- Because a dividend on common stock is non-tax-deductible, we say it cost 100% of amount paid. From a purely corporate cash flow point of view, the firm would be indifferent between paying $100,000 in interest and $65,000 in dividends.

LESSON 6

DEPRECATION AS A TAX SHIELD

- Although depreciation is not a new source of funds, it provides the important function of shielding part of our income from taxes.
- Again, we take the same two corporations – one charges off $100,000 in depreciation, other charges off none.

	Corporation A	Corporation B
Earnings before interest and taxes	$400,000	$400,000
- Depreciation	**100,000**	**0**
Earnings before taxes (taxable income)	300,000	400,000
- Taxes @35%	105,000	140,000
Earnings after taxes	$195,000	$260,000
+ Dep. charged without cash outlay	**100,000**	**0**
Cash flow	$295,000	$260,000
Difference	**$35,000**	

It can also be computed as: Depreciation x Tax rate
= $100,000 x 35%
= $35,000

Highlighted by MR Shoaib

Depreciation as a Tax Shield

- Corporation A enjoys $35,000 more in cash flow, since depreciation shielded $100,000 from taxation in Corporation A and saved $35,000 in taxes, which eventually appeared in cash flows.

Financial Cash Flow

- In finance, the most important item that can be extracted from financial statements is the actual cash flow of the firm.
- Since there is no magic in finance, it must be the case that the cash received from the firm's assets must equal the cash flows to the firm's creditors and stockholders.

Cash Flow from Assets = Cash Flow to Creditors + Cash Flow to Stockholders

Cash Flow from Assets

- Cash flow from assets involves three components
  - Operating Cash flow
  - Capital Spending
  - Change in Net Working capital

Operating Cash Flow

- This refers to the cash flow that results from the firm's day-to-day activities of producing and selling
- Expenses related to firm's financing of its assets are not included since they are not operating expenses
- To calculate OCF, we calculate revenues minus costs (including taxes being paid in cash), but don't include:
  - Depreciation since it is not a cash out flow
  - Interest because it is a financing expense

- OCF is very significant as it tells whether or not the firm's cash inflows from its business operations are sufficient to cover its everyday cash outflows.
- So, a negative operating cash flow is a sign of trouble.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Operating Cash Flow	
EBIT	$219
Depreciation	90
Current Taxes	(71)
OCF	**$238**

Capital Spending

- Some portion of the firm's cash flow is reinvested in the firm. Capital spending refers to the net spending on fixed assets (Purchases of fixed assets less sale of fixed assets).
- Net Capital Spending could be negative if the firm sold off more assets than it purchased.
- Depreciation of the respective assets is accounted for in this regard.

Highlighted by MR Shafiq

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Capital Spending	
Purchase of fixed assets	$(198)
Sales of fixed assets	25
Capital Spending	**$(173)**

$173= $149 + $24 (increase in property, plant, and equipment + increase in intangible assets)

Change in Net Working Capital

- It is the amount spent on Net Working Capital, and represents the net increase in current assets over current liabilities.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

NWC grew to $275 million in 20X2 from $252 million in 20X1. This increase of $23 million is the addition to NWC.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Total Cash Flow of the Firm

Highlighted By MR Shahzad

Cash Flow to Creditors

- Cash flow to creditors is calculated as interest paid less net new borrowing.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Cash Flow to Creditors	
Interest	$(49)
Retirement of debt	(73)
Debt service	$(122)
Proceeds from new debt sales	86
Total	**$ (36)**

Cash Flow to Stockholders

- Cash flow to stockholders is calculated as dividends paid less net new equity raised.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Cash Flow to Stockholders	
Dividends	$(43)
Repurchase of Stock	(6)
Cash to Stockholders	$(49)
Proceeds from new stock issue	43
Total	**$ (6)**

Highlighted by MR Shahzad

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

Total Cash Flow to Creditors and Stockholders.

XYZ CORPORATION Financial Cash Flow 20X2 (in $ millions)	
Cash Flow of the Firm	
Operating cash flow	$238
Capital spending	(173)
Additions to net working capital	(23)
Total	**$42**
Cash Flow of Investors in the Firm	
Debt	$36
Equity	6
Total	**$42**

The cash from received from the firm's assets must equal the cash flows to the firm's creditors and stockholders:

CF (Assets)

=

CF (Creditors) + CF (Stockholders)

Cash Flow Summary

Cash Flow identity:

- Cash flow from Assets = Cash Flow to creditors + Cash flow to Stockholders

- Cash flow from Assets:

  Cash flow from assets = Operating Cash Flow
  - Net Capital Spending
  - Change in Net Working Capital

  Where,
  Operating cash flow = Earnings before Interest and taxes + Depreciation – Taxes
  Net Capital Spending = Ending Net Fixed Assets - Beginning Net Fixed Assets + Depreciation
  Change in NWC = Ending NWC – Beginning NWC

- Cash flow to creditors (bondholders):

  Cash flow to creditors = Interest paid – Net new borrowings

- Cash flow to stockholders (owners):

  Cash flow to stockholders = Dividends Paid – Net new equity raised

Highlighted by MR Shahbaz

LESSON 7

THE STATEMENT OF CASH FLOWS

- There is an official accounting statement called the statement of cash flows.
- This helps explain the change in accounting cash,
- The three components of the statement of cash flows are
  - Cash flow from operating activities
  - Cash flow from investing activities
  - Cash flow from financing activities

The Balance Sheet of the XYZ Corporation

XYZ CORPORATION Balance Sheet 20X2 and 20X1 (in $ millions)		
Assets	**20X2**	**20X1**
Current assets:		
Cash and equivalents	$140	$107
Accounts receivable	294	270
Inventories	269	280
Other	58	50
Total current assets	**$761**	**$707**
Fixed assets:		
Property, plant, and equipment	$1423	$1274
Less accumulated depreciation	- 550	- 460
Net property, plant, and equipment	873	814
Intangible assets and other	245	221
Total fixed assets	$1,118	$1,035
Total assets	**$1,879**	**$1,742**

XYZ CORPORATION Balance Sheet 20X2 and 20X1 (in $ millions)		
Liabilities (Debt) and Stockholder's Equity	**20X2**	**20X1**
Current Liabilities:		
Accounts payable	$213	$197
Notes payable	50	53
Accrued expenses	223	205
Total current liabilities	$486	$455
Long-term liabilities:		
Deferred taxes	$117	$104
Long-term debt	471	458
Total long-term liabilities	$588	$562
Stockholder's equity:		
Preferred stock	$39	$39
Common stock ($1 per value)	55	32
Capital surplus	347	327
Accumulated retained earnings	390	347
Less treasury stock	-26	-20
Total equity	$805	$725
Total liabilities and stockholder's equity	**$1,879**	**$1,742**

Highlighted by MR Shahbaz

Notes:

1. Long-term debt rose by ($471 million–$458 million) $13 million. This is the difference between $86 million new debt and $73 million in retirement of old debt.

2. Treasury stock rose by $6 million. This reflects the repurchase of $6 million stock.

3. XYZ Corporation reports $43 million in new equity. The company issued 23 million shares at a price of $1.87. The par value of common stock is increased by $23 million and capital surplus is increased by $20 million.

Income Statement of XYZ Corporation

XYZ CORPORATION Income Statement 20X2 (in $ millions)	
Total operating revenues	$2,262
Cost of goods sold	-1655
Selling, general, and administrative expenses	-327
Depreciation	-90
Operating income	$190
Other income	29
Earnings before interest and taxes	$219
Interest expense	-49
Pretax income	$170
Taxes	-84
Current: $71	
Deferred: $13	
Net income	$86
Retained earnings:	$43
Dividends:	$43

Highlighted by MR Shahbaz

Notes:

1. There are 29 million shares outstanding. Earnings per share and dividends per share can be calculated as follows:

$$\textit{Earnings per share} = \frac{\textit{Net Income}}{\textit{Total share outstanding}}$$

$$\textit{Earnings per share} = \frac{\$86}{\$29}$$

Earnings per share = $ 2.97 per share

$$\textit{Dividend per share} = \frac{\textit{Dividend}}{\textit{Total share outstanding}}$$

$$\textit{Dividend per share} = \frac{\$43}{\$29}$$

Dividend per share = $ 1.48 per share

Cash Flow from Operating Activities of XYZ Corporation

To calculate cash flow from operations, start with net income, add back noncash items like depreciation and adjust for changes in current assets and liabilities (other than cash).

XYZ CORPORATION Cash Flow from Operating Activities 20X2 (in $ millions)	
Operations Net Income	**$86**
Depreciation	90
Deferred Taxes	13
Changes in Assets and Liabilities	
Accounts Receivable	(24)
Inventories	11
Accounts Payable	16
Accrued Expenses	18
Notes Payable	(3)
Other	(8)
Total Cash Flow from Operations	**$199**

Cash Flow from Investing Activities of XYZ Corporation

Cash flow from investing activities involves changes in capital assets: acquisition of fixed assets and sales of fixed assets (i.e. net capital expenditures).

XYZ CORPORATION Cash Flow from Investing Activities 20X2 (in $ millions)	
Acquisition of fixed assets	**$(198)**
Sales of fixed assets	25
Total Cash Flow from Investing Activities	**$(173)**

Highlighted by MR Shahbaz

Cash Flow from Financing Activities of XYZ Corporation

Cash flows to and from creditors and owners include changes in equity and debt.

XYZ CORPORATION Cash Flow from Financing Activities 20X2 (in $ millions)	
Retirement of debt (includes notes)	**$(73)**
Proceeds from long-term debt sales	86
Dividends	(43)
Repurchase of stock	(6)
Proceeds from new stock issue	43
Total Cash Flow from Financing Activities	**$ 7**

Cash Flow Statement of XYZ Corporation

The statement of cash flows is the addition of cash flows from operations, cash flows from investing activities, and cash flows from financing activities.

XYZ CORPORATION Statement of Cash Flows 20X2 (in $ millions)	
Operations	
Net Income	$86
Depreciation	90
Deferred Taxes	13
Changes in Assets and Liabilities	
Accounts Receivable	(24)
Inventories	11
Accounts Payable	16
Accrued Expenses	18
Notes Payable	(3)
Other	(8)
Total Cash Flow from Operations	**$199**
Investing Activities	
Acquisition of fixed assets	$(198)
Sales of fixed assets	25
Total Cash Flow from Investing Activities	**$(173)**
Financing Activities	
Retirement of debt (includes notes)	$(73)
Proceeds from long-term debt sales	86
Dividends	(43)
Repurchase of stock	(6)
Proceeds from new stock issue	43
Total Cash Flow from Financing	**$7**
Change in Cash (on the balance sheet)	**$33**

Highlighted by MR Shahbaz

Significance of Financial Statements

- A good Working Knowledge of financial Statements is desirable simply because these statements are the primary means of communicating financial information both within and outside the firm

External Uses of Statement Analysis

- ***Trade Creditors*** -- Focus on the liquidity of the firm.
- ***Bondholders*** -- Focus on the long-term cash flow of the firm.
- ***Shareholders*** -- Focus on the profitability and long-term health of the firm.

Internal Uses of Statement Analysis

- ***Plan*** -- Focus on assessing the current financial position and evaluating potential firm opportunities.

- ***Control*** -- Focus on return on investment for various assets and asset efficiency.
- ***Understand*** -- Focus on understanding how suppliers of funds analyze the firm.

Significance of Financial Statements

- The reason, we rely on accounting figures for much of our financial information is that we are almost always unable to obtain all of market information we want.
- The only meaningful yardstick for evaluation business decisions is whether or not they create economic value.
- Clearly, one important goal of the accountant is to report financial information to the user in a form useful for decision making.
- But the financial statements don't come with a user's guide.
- We will try to fill up this gap through learning a comprehensive analysis of financial statements.

Standardized Financial Statements

- One obvious thing we want to do with a company's financial statements is to compare them to those of other.
- It is almost impossible to directly compare the financial statements for two companies because of differences in size. So we will try to standardize the financial statements.

Common-Size Statements

- One very common and useful way of standardized comparison is to work with percentages instead of dollars.
- So, a standardized financial statement presenting all items in percentages is called a common-size statement.
- Balance sheet items are shown as a percentage of total assets and income statement items as a percentage of sales.

Highlighted by MR Shahbaz

LESSON 8

COMMON SIZE STATEMENTS

- One very common and useful way of standardized comparison is to work with percentages instead of dollars.
- So, a standardized financial statement presenting all items in percentages is called a common-size statement.
- Balance sheet items are shown as a percentage of total assets and income statement items as a percentage of sales.

A2Z Inc.
Balance Sheet as of December 31
($ in millions)

Assets	**20X1**	**20X2**
Current Asset		
Cash	$84	$98
Accounts Receivable	165	188
Inventory	393	422
Total Current Assets	642	708
Fixed Assets		
Net Plant and Equipment	2,731	2,880
Total Assets	$3,373	$3,588
Liabilities and Equity	**20X1**	**20X2**
Current Liabilities		
Accounts Payable	312	344
Notes Payable	231	196
Total Current Liabilities	543	540
Long-term Debts	531	457
Stockholders' Equity		
Common Stock and Paid in surplus	500	550
Retained Earning	1,799	2,041
Total Stockholders' Equity	2,299	2,591
Total Liabilities and Equity	$3,373	$3,588

Highlighted by MR Shahzad

A2Z Inc.,
Common-Size Balance Sheet

Assets	20X1	20X2
Current Asset		
Cash	2.5%	2.7%
Accounts Receivable	4.9%	5.2%
Inventory	11.7%	11.8%
Total Current Assets	19.1%	19.7%
Fixed Assets		
Net Plant and Equipment	80.9%	80.3%
Total Assets	100.0%	100.0%
Liabilities and Equity	**20X1**	**20X2**
Current Liabilities		
Accounts Payable	9.2%	9.6%
Notes Payable	6.8%	5.5%
Total Current Liabilities	16.0%	15.1%
Long-term Debts	15.7%	12.7%
Stockholders' Equity		
Common Stock and Paid in surplus	14.8%	15.3%
Retained Earning	53.3%	56.9%
Total Stockholders' Equity	68.1%	72.2%
Total Liabilities and Equity	100.0%	100.0%

Highlighted by MR Shahbaz

A2Z Inc., Common-Size Balance Sheet

More on Standardized Statements

- Suppose we ask: "What happened to A2Z's net plant and equipment (NP&E) over the period?"
- Based on the 20X1 and 20X2 B/S, NP&E rose from $2,731 to $2,880, so NP&E rose by $149.
- Did the firm's NP&E go up or down? Obviously, it went up, but so did total assets. In fact, looking at the standardized statements, NP&E went from 80.9% of total assets to 80.3% of total assets.

A2Z Inc., Common-Size Balance Sheet

More on Standardized Statements

- If we standardized the 20X2 numbers by dividing each by the 20X1 number, we get a common base year statement. In this case, $2,880 / $2,731 = 1.0545, so NP&E rose by 5.45% over this period.
- If we standardized the 20X2 common size numbers by dividing each by the 20X1 common size number, we get a combined common size, common base year statement. In this case, 80.3%/

80.9% = 99.26%, so NP&E almost remained the same as a percentage of assets.

- In absolute terms, NP&E is up by $149 or 5.45%, but relative to total assets, NP&E fell by 0.6%.

A2Z Inc., Common-Size Balance Sheet

More on Standardized Statements

- Current assets rose from 19.1% in 20X1 to 19.7% in 20X2
- Current liabilities declined from 16.0% to 15.1% of total liabilities and equity over the same time.
- Total equity rose from 68.1% of total liabilities and equity to 72.2%.
- Overall, A2Z's liquidity as measured by current assets compared to current liabilities, increased over the year. Also, A2Z's indebtness diminished as a percentage of total assets.
- So we may conclude that balance sheet as grown stronger.

A2Z Inc., Income Statement
For the Year 20X2
($ in millions)

Net Sales		$2311
Cost of Goods Sold		1344
Depreciation		276
Earnings before interest and taxes	691	
Interest	141	
Taxable Income		550
Taxes		187
Net Income		$363
Dividends	121	
Retained Earnings	242	

Highlighted by MR Shahbaz

A2Z Inc., Common-Size Income Statement

Net Sales		100.0%
Cost of Goods Sold		58.2%
Depreciation		11.9%
Earnings before interest and taxes	29.9%	
Interest	6.1%	
Taxable Income		23.8%
Taxes		8.1%
Net Income		15.7%
Dividends	5.2%	
Retained Earnings	10.5%	

A2Z Inc., Common-Size Income Statement

- This statement tells us what happened to each dollar in sales.
- For A2Z interest expense eats up 6.1% of sales, while taxes take another 8.1% of sales figure.
- Following this, 15.7% of revenues from sales flow down to bottom as net income; one-third of which is paid in dividends and remainder two-thirds is taken as retained earnings for business.
- As far as cost is concerned, 58.2% of revenues are spent on the goods sold

Standardized Financial Statements

- Although an organization's common-size statements provide a better analytical insight into its strength and standing, yet its performance and efficiency can be better judged by comparing these with those of the firm's competitors.

Ratio Analysis

- Another way of avoiding the problems involved in comparing companies of different sizes, is to calculate and compare financial ratios.
- One problem with ratios is that different people and different sources frequently don't compute them in exactly the same way.
- While using ratios as a tool for analysis, you should be careful to document how you calculate each one, and, if you are comparing your numbers to those of another source, be sure you know how their numbers are computed.

Ratio Analysis

- For each of the ratios we discuss, several questions come to mind:
  - How is it computed?
  - What is it intended to measure, and why might we be interested?
  - What is the unit of measurement?
  - What might a high or low value be telling? How might such values be misleading?
  - How could this measure be improved?

Ratio Analysis

- Financial ratios are traditionally grouped into the following categories:
  - **Short-term solvency, or liquidity, ratios**
    - Ability to pay bills in the short-run
  - **Long-term solvency, or financial leverage, ratios**
    - Ability to meet long-term obligations
  - **Asset management, or turnover, ratios**
    - Intensity and efficiency of asset use
  - **Profitability ratios**
    - Ability to control expenses
  - **Market value ratios**
    - Going beyond financial statements

Short-Term Solvency, or Liquidity, Measures

- The primary concern, to which these ratios relate, is the firm's ability to pay its bills over the short run without undue stress. So these ratios focus on current assets and current liabilities.
- Liquidity ratios are particularly interesting to short-term creditors. Since financial managers are constantly working with banks and other short-term lenders, an understanding of these ratios is essential

- Current assets and liabilities
  - Their book values and market values are likely to be similar.
  - They can and do change fairly rapidly, hence unpredictable

Highlighted by Mr Shahzad

Current Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- Because current assets and liabilities are converted into cash over the following 12 months, the current ratio is a measure of short run liquidity.
- The unit of measurement is either dollars or times.

Current Ratio

- For A2Z Corporation, the 20X2 current ratio is:

$$\text{Current Ratio} = \frac{\$708}{\$540} = 1.31 \text{ times}$$

- We can say that
  - A2Z has a $1.31 in current assets for every $1 in current liabilities OR
  - A2Z has its current liabilities covered 1.31 times over.
- To a creditor (particularly a short-term creditor like supplier), the higher the current ratio, the better
- To firm, high current ratio indicates liquidity, but it may also indicate an inefficient use of cash and other short-term assets.
- We would expect to see a current ratio of at least 1, because a current ratio of less than 1 would mean that net working capital is negative
- Like any other ratio, current ratio is affected by various transactions.
  - If a firm borrows over long-term,
    - The short run effect would be an increase in cash as well as in long term liabilities.
    - Current liabilities would not be affected, so the current ratio would rise.
- An apparently low current ratio may not be a bad sign for a company with a large reserve of unlimited borrowing power.

Highlighted by Mir Shahbaz

Current Events

- A firm wants to pay-off some of its suppliers and creditors. What would happen to current ratio?
  - Current ratio moves away from 1. if it is greater than 1 it will get bigger. But if it is less than 1, it will get smaller.
- Suppose a firm has $4 in current assets and $2 in current liabilities for a current ratio of 2. and uses $1 in cash to reduce current liabilities, then new current ratio is ($4-1) / ($2-1) = 3
- Reversing the situation to $2 in current assets and $4 in current liabilities, the change will cause current ratio to fall to 1/3 from 1/2.
- Suppose a firm buys some inventory. What would happen in this case?
  - Nothing happens to current ratio. Because in this scenario, one current asset (cash) goes down while another current asset (inventory) goes up. Total current assets are unaffected.
- What happens if a firm sells some merchandise?
  - Current ratio would usually rise because inventory is shown at cost and sale would normally be at something greater than cost (difference is markup).
- So, the increase in either cash or receivables is greater than the decrease in inventory.
- This increases current assets and current ratio rises.

Quick (or Acid-Test) Ratio

- Inventory is often the least liquid current asset. And its book values are least reliable as measures of market value since the quality of inventory isn't considered. Some of the inventory may turn out to be damaged, obsolete or lost.
- Relatively large inventories are often a sign of short-term trouble.
- The firm may have overestimated sales and overbought or overproduced as a result, hence tied up a substantial portion of its liquidity in slow moving inventory.

$$\text{Quick Ratio} = \frac{\text{Current Assets - Inventory}}{\text{Current Liabilities}}$$

- For A2Z, this ratio in 20X2 was:

$$\text{Quick Ratio} = \frac{\$708 - \$\,422}{\$\,540} = 0.53 \text{ times}$$

The quick ratio here tells a somewhat different story than the current ratio, because inventory accounts for more than half of A2Z's current assets.

- If the same figure is for an aircraft manufacturing corporation, then this would certainly be cause for a BIG concern.

Cash Ratio

- A very short-term creditor may be interested in the cash ratio.

$$\text{Cash Ratio} = \frac{\text{Cash}}{\text{Current Liabilities}}$$

- Current ratio for A2Z in 20X2 was 0.18

Highlighted by MR Shahbaz

LESSON 9

RATIO ANALYSIS

A2Z Inc., Balance Sheet
As of December 31
($ in millions)

Assets	**20X1**	**20X2**
Current Asset		
Cash	$84	$98
Accounts Receivable	165	188
Inventory	393	422
Total Current Assets	$642	$708
Fixed Assets		
Net Plant and Equipment	$2,731	$2,880
Total Assets	$3,373	$3,588
Liabilities and Equity	**20X1**	**20X2**
Current Liabilities		
Accounts Payable	$312	$344
Notes Payable	231	196
Total Current Liabilities	$543	$540
Long-term Debts	531	457
Stockholders' Equity		
Common Stock and Paid in surplus	$500	$550
Retained Earning	1,799	2,041
Total Stockholders' Equity	2,299	2,591
Total Liabilities and Equity	$3,373	$3,588

Highlighted by MR Shahbaz

A2Z Inc., Income Statement
For the Year 20X2
($ in millions)

Net Sales		$2,311
Cost of Goods Sold		1,344
Depreciation		276
Earnings before interest and taxes	$691	
Interest	141	
Taxable Income		550
Taxes		187
Net Income		$363
Dividends	$121	
Retained Earnings	$242	

Current Ratio

Current Events

- Suppose a firm buys some inventory. What would happen in this case?
- Nothing happens to current ratio. Because in this scenario, one current asset (cash) goes down while another current asset (inventory) goes up. Total current assets are unaffected.

What happens if a firm sells some merchandise?

- Current ratio would usually rise because inventory is shown at cost and sale would normally be at something greater than cost (difference is markup).
- So, the increase in either cash or receivables is greater than the decrease in inventory.
- This increases current assets and current ratio rises.

Current Ratio

- A firm wants to pay-off some of its suppliers and creditors. What would happen to current ratio?
- Current ratio moves away from 1. if it is greater than 1 it will get bigger. But if it is less than 1, it will get smaller.
- Suppose a firm has $4 in current assets and $2 in current liabilities for a current ratio of 2. and uses $1 in cash to reduce current liabilities, then new current ratio is ($4-1) / ($2-1) = 3
- Reversing the situation to $2 in current assets and $4 in current liabilities, the change will cause current ratio to fall to 1/3 from 1/2

Highlighted by MR Sadaqat

Long Term Solvency Measures

- These ratios are intended to address the firm's long-run ability to meet its obligations, or its financial leverage.

Total Debt Ratio

This ratio takes into account all debts of all maturities to all creditors. It is computed as:

$$\text{Total Debt Ratio} = \frac{\text{Total Assets - Total Equity}}{\text{Total Assets}}$$

For A2Z Corporation,

$$\text{Total Debt Ratio} = \frac{\$3{,}588 - \$\,2{,}591}{\$3{,}588} = 0.28 \text{ times}$$

- So, A2Z uses 28% debt. Whether this is high or low or whether it even makes any difference depends on whether or not capital structure matters.
- A2Z has 28% debt against total assets, thus there is 72% equity against total assets.
- Here we draw two variations out of total debt ratio
- Debt-equity ratio
- Equity multiplier
- Debt–Equity ratio = Total Debt / Total Equity
  = 28% / 72% = 0.39 times
- Equity Multiplier = Total Assets / Total Equity
  = 100% / 72% = 1.39 times
  OR
  = 1 + Debt-Equity ratio = 1.39 times

Interest Coverage Ratio

- Also known as Times Interest Earned (TIE) ratio, refers to the ability of the firm to cover is interest obligations.

$$\text{Interest Coverage Ratio} = \frac{\text{Earning Befor Interest and Taxes}}{\text{Interest}}$$

For A2Z corporation,

$$\text{Interest Coverage Ratio} = \frac{\$\ 691}{\$141} = 4.9 \text{ times}$$

Cash Coverage Ratio

- A problem with Interest Coverage Ratio is that it is based on Earnings before Interest and Taxes (EBIT) which is not really a measure of cash available to pay interest.
- The reason is that depreciation, a non-cash expense has been deducted out. So we use:

$$\text{Cash Coverage Ratio} = \frac{\text{EBIT+Depreciation}}{\text{Interest}}$$

For A2Z corporation,

$$\text{Cash Coverage Ratio} = \frac{\$691 + \$\ 276}{\$141}$$

$$\text{Cash Coverage Ratio} = \frac{\$\ 967}{\$141} = 6.9 \text{ times}$$

Highlighted by MR Shahzaz

Asset Management or Turnover Measures

The measures in this section are sometimes called Asset Utilization Ratios. These are intended to describe how efficiently or intensively a firm uses its assets to generate sales.

Inventory Turnover Ratio

Inventory turnover can be calculated as:

$$\text{Inventory turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Inventory}}$$

For A2Z corporation,

$$\text{Inventory turnover Ratio} = \frac{\$\ 1{,}344}{\$422} = 3.2 \text{ times}$$

So, A2Z sold off or turned over the entire inventory 3.2 times. As long as stock-out and foregoing sales situation doesn't arise, the higher this ratio is, the more efficiently inventory is being managed.

Days' Sales in Inventory

If we know sales were turned over 3.2 times during the year, we can calculate easily how long it took to turnover on average.

$$\text{Day's sales in Inventory} = \frac{365 \text{ days}}{\text{Inventory Turnover}}$$

For A2Z corporation,

$$\text{Day's sales in Inventory} = \frac{365}{3.2} = 114 \text{ days}$$

So, inventory stays for just less than 4 months before being sold or it would take 114 days to sell off current inventory.

Receivables Turnover

Now we take a look on how fast we collect on the sales of inventory.

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts Receivables}}$$

For A2Z corporation,

$$\text{Receivables turnover} = \frac{\$\,2{,}311}{\$\,188} = 12.3 \text{ times}$$

So A2Z collected its outstanding credit accounts and re-loaned the money 12.3 times during the year. *(Assuming all the sales are credit sales. If not, we use only credit sales for this ratio)*

Days' Sales in Receivables

$$\text{Day's Sales in Receivables} = \frac{365 \text{ days}}{\text{Receivable Turnover}}$$

For A2Z corporation,

$$\text{Day's Sales in Receivables} = \frac{365}{12.3} = 30 \text{ days}$$

So A2Z collects on its credit sales in a month, or the firm has 30 days' worth of sales uncollected. This ratio is also called **Average Collection Period**.

A Variation: Payables Turnover

It describes a how long does the firm take to pay its bills, and is computed as:

$$\text{Payables Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Accounts payables}}$$

$$\text{Payables Turnover} = \frac{\$\,1{,}344}{\$344} = 3.9 \text{ times}$$

So, days it took to turn-over the payables are:

$$= \frac{365}{3.9} = 94 \text{ days}$$

This figure is very significant to the current as well as potential creditors of A2Z.

Highlighted by Mr Shahzad

LESSON 10

RATIO ANALYSIS (CONT…)

A2Z Inc., Balance Sheet
As of December 31
($ in millions)

Assets	**20X1**	**20X2**
Current Asset		
Cash	$84	$98
Accounts Receivable	165	188
Inventory	393	422
Total Current Assets	$642	$708
Fixed Assets		
Net Plant and Equipment	$2,731	$2,880
Total Assets	$3,373	$3,588
Liabilities and Equity	**20X1**	**20X2**
Current Liabilities		
Accounts Payable	$312	$344
Notes Payable	231	196
Total Current Liabilities	$543	$540
Long-term Debts	531	457
Stockholders' Equity		
Common Stock and Paid in surplus	$500	$550
Retained Earning	1,799	2,041
Total Stockholders' Equity	2,299	2,591
Total Liabilities and Equity	$3,373	$3,588

Highlighted by MR Shahbaz

A2Z Inc., Income Statement
For the Year 20X2
($ in millions)

Net Sales		$2,311
Cost of Goods Sold		1,344
Depreciation		276
Earnings before interest and taxes	$691	
Interest	141	
Taxable Income		550
Taxes		187
Net Income		$363
Dividends	$121	
Retained Earnings	$242	

Total Asset turnover

$$\text{Total Assets turnover} = \frac{\text{Sales}}{\text{Total Assets}}$$

For A2Z corporation,

$$\text{Total Assets turnover} = \frac{\$2{,}311}{\$3{,}558} = 0.64 \text{ times}$$

In other words, for every dollar in assets, A2Z generated $0.64 in sales.

Capital Intensity Ratio

It is simply the reciprocal of total assets turnover, or:

$$\text{Capital Intensity Ratio} = \frac{\text{Total Assets}}{\text{Sales}}$$

- It is interpreted as the dollar investment in assets needed to generate $1 in sales. Higher values represent capital intensive industries.
- For A2Z, this ratio is computed to be 1.56, i.e. A2z has to invest $1.56 in assets to get $1 in sales.

Profitability Measures

- In one form or the other, these ratios are intended to measure how efficiently the firm uses its assets and how efficiently the firm manages its operations.
- The focus in this group is on the bottom line – net income.

Profit Margin

Every company, big or small, pays very close attention to their profit margin.

$$\text{Profit Margin Ratio} = \frac{\text{Net Income}}{\text{Sales}}$$

For A2Z Company,

$$\text{Profit Margin Ratio} = \frac{\$\,363}{\$2{,}311} = 15.7\ \%$$

- So in accounting sense, A2Z generates a little less than 16 cents in profit for every dollar in sales.
- Other Things being equal, a relatively high profit margin is obviously desirable, corresponding to low expenses vs. sales.
- But Other Things are not always equal!
- For example, lowering the sales price will usually increase unit sales but will normally cause profit margin to shrink. Total Profit of operating cash flow may go up or down.

Return on Assets

Return on Assets (ROA) is a measure of profit per dollar of assets:

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Total Assets}}$$

Highlighted by MR Shahzaz

For A2Z,

$$\text{Return on Assets} = \frac{\$\,363}{\$3{,}588} = 10.12\%$$

Return on Equity

- Return on equity (ROE) is a measure of how the stockholders fared during the year.
- Since benefiting the shareholders is the goal of corporation, ROE is a true bottom line measure of performance.

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Total Equity}}$$

For A2Z,

$$\text{Return on Equity} = \frac{\$\,363}{\$2{,}591} = 14\%$$

Therefore, for every dollar in equity, A2Z generated 14 cents in profit. But this is correct in accounting terms only.

ROA and ROE

- Because ROA and ROE are most widely used and commonly cited numbers, so it should be kept in mind that these are accounting rates of return.
- That is why these are called return on book assets and return on book equity
- ROE is sometimes called return on Net Worth

Market Value Measures

- This group of measures is based, in part, on information not necessarily contained in financial statements, like market price per share.
- These measures can be calculated directly only for publicly traded companies.

Earnings per Share

Assuming,

- A2Zhas 33 million shares outstanding and
- Stock sold for $88 per share at the end of year.

So, the Earnings per Share (EPS) will be:

$$\text{Earning Per Share} = \frac{\text{Net Income}}{\text{Shares Outstanding}} = \frac{\$\,363}{33} = \$11$$

Price-Earning Ratio

Price-earnings or PE ratio is defined as:

$$\text{Price-Earning Ratio} = \frac{\text{Price Per Share}}{\text{Earnings per share}}$$

For A2Z,

$$\text{Price-Earning Ratio} = \frac{\$\,88}{\$\,11} = 8 \text{ times}$$

- So A2Z shares sell for eight times earnings or it carries a PE multiple of 8.

Highlighted by Mir Shahzad

- Since PE ratio measures how much investors are willing to pay per dollar of current earnings, higher PEs are often taken to mean that the firm has significant prospects for future growth.
- If a firm had no or almost no earnings, its PE would probably be quite large; so careful interpretation is required.

Book Value per share

Book Value is calculated as:

$$\text{Book Value} = \frac{\text{Total equity}}{\text{No.of shares outstanding}}$$

For A2Z,

$$\text{Book Value} = \frac{\$\ 2{,}591}{33} = \$78.5$$

Since book value per share is an accounting number, it reflects historical costs.

Market-to-Book ratio

It is defined as:

$$\text{Mark-to-Book Ratio} = \frac{\text{Market value per share}}{\text{Book value per share}}$$

For A2Z,

$$\text{Mark-to-Book Ratio} = \frac{\$\ 88}{\$78.5} = 1.12 \text{ times}$$

- The market-to-Book ratio compares the market value of the firm's investment to their costs.
- A value less than 1 could mean that the firm has not been successful overall in creating value for its stockholders.

The Du Pont Identity

- The difference between the two profitability measures, ROA and ROE, is the use of debt financing, or financial leverage.
- The relationship between these measures can be illustrated by decomposing ROE into its component parts.

Recall,

$$\text{ROE} = \frac{\text{Net Income}}{\text{Total Equity}}$$

Multiplying it by Assets / Assets (without changing anything)

$$\text{ROE} = \frac{\text{Net Income}}{\text{Total Equity}} = \frac{\text{Net Income}}{\text{Total Equity}} \times \frac{Assets}{Assets}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Assets}} \times \frac{Assets}{Total\ Equity}$$

So, we have expressed ROE as a product of two other ratios; ROA and the equity multiplier.

ROE = ROA × Equity Multiplier
ROE = ROA × (1 + Debt-Equity Ratio)

Highlighted by MR Shahzad

LESSON 11

THE DU-PONT IDENTITY

- The difference between the two profitability measures, ROA and ROE, is the use of debt financing, or financial leverage.
- The relationship between these measures can be illustrated by decomposing ROE into its component parts.

Recall,

$$\text{ROE} = \frac{\text{Net Income}}{\text{Total Equity}}$$

Multiplying it by Assets / Assets (without changing anything)

$$\text{ROE} = \frac{\text{Net Income}}{\text{Total Equity}} = \frac{\text{Net Income}}{\text{Total Equity}} \times \frac{Assets}{Assets}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Assets}} \times \frac{Assets}{Total\ Equity}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Assets}} \times \frac{Assets}{Total\ Equity}$$

So, we have expressed ROE as a product of two other ratios; ROA and the equity multiplier

ROE = ROA × Equity Multiplier
ROE = ROA × (1 + Debt-Equity Ratio)

- Looking back at A2Z:
- Debt-Equity Ratio = 0.39
- ROA = 10.12%
- while ROE calculated previously = 14%
- Now, using the decomposition method:

ROE = 10.12% × 1.39 = 14%

We can further decompose ROE by multiplying the top and bottom by total sale:

$$\text{ROE} = \frac{\text{Sales}}{\text{Sales}} \times \frac{\text{Net Income}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Total Equity}}$$

Rearranging a bit,

$$\text{ROE} = \frac{Net\ Income}{Sales} \times \frac{Sales}{Assets} \times \frac{Assets}{Total\ Equity}$$

Return on Assets

ROE = Profit Margin × Total Assets Turnover × Equity Multiplier

This last Expression is called **Du Pont Identity** after the Du Pont Corporation, which popularized its use.

Highlighted by MR Shahbaz

ROE = 15.7% × 0.64 × 1.39
ROE = 14%

- The Du Pont identity tells us that ROE is affected by three things:
  - Operating efficiency (as measured by profit margin)
  - Asset use efficiency (as measured by total assets turnover)
  - Financial Leverage (as measured by equity multiplier)

Considering the Du Pont identity,

- It appears that a firm could leverage up its ROE by increasing its amount of debt
- This will only happen if ROA exceeds interest rate on debt.
- The decomposition of ROE is a convenient way of systematically approaching the financial statements analysis.
- If ROE is unsatisfactory by some measure, then Du Pont identity tells you where to start looking for the reasons.

Dividend Payout

We have seen earlier that the net income is divided into two pieces.

- The first piece is cash dividend paid to stockholders
- Leftover amount is the addition to retained earnings
- A2Z's Net Income was $363, of which $121 was paid out in dividends. As a percentage:

$$\text{Dividend Payout ratio} = \frac{\textit{Cash Dividends}}{\text{Net Income}}$$

$$\text{Dividend Payout ratio} = \frac{\$121}{\$363} = 33\frac{1}{3}\ \%$$

Retention Ratio

Anything A2Z does not pay out in form of dividends must be retained in the firm. So retention ratio is:

$$\text{Retention ratio} = \frac{\text{Retained Earnings}}{\text{Net Income}}$$

$$\text{Retention ratio} = \frac{\$242}{\$363} = 66\frac{2}{3}\ \%$$

- So, A2Z retains two-thirds of its net income.
- The retention ratio is also known as the plowback ratio, as this is the amount which is plowed back into the business.

Payout and Retention

Q: LMN Corporation pays out 40% of net income in form of dividends. What is its retention ratio?
A: If payout ratio is 40%, retention ratio is: 1 – 40% = 60%

Q: If net income of LMN is $800, how much did stockholders actually receive?
A: Dividends are $800 x 40% = $320

Internal and Sustainable Growth

- Firm's Return on Assets and Return on Equity are frequently used to calculate two additional numbers, both of which have to do with the firm's ability to grow.

Highlighted by MR Shahzad

- Investors and others are frequently interested in knowing how rapidly a firm's sales can grow.
- But the important thing to recognize is that if sales are to grow, assets have to grow as well, at least over the long run.
- Further if assets are to grow, then the firm must somehow obtain money to pay for the purchases.
- So, the growth has to be financed. And more so, a firm's ability to grow depends on its financing policies.
- A firm has two broad sources of financing:
  - Internal financing simply refers to what the firm earns and subsequently plows back into the business.
  - External financing refers to funds raised by either borrowing money or selling stock.

Internal Growth Rate

A firm having a policy of internal financing, won't borrow funds and won't sell any new stock. Internal growth rate represents how rapidly the firm grows.

$$\text{Internal Growth Rate} = \frac{ROA \times b}{1-(ROA \times b)}$$

Where ROA is return on assets and b is the retention ratio

For A2Z this rate is:

$$\text{Internal Growth Rate} = \frac{0.1012 \times \frac{2}{3}}{1-\left(0.1012 \times \frac{2}{3}\right)}$$

Internal Growth Rate = 7.23%

Highlighted by MR Shahzad

Sustainable Growth Rate

- If a firm only relies on the internal financing, then through time, its total debt ratio will decline, because assets will grow but total debt will remain the same (or even fall if some is paid off).
- Frequently, firms have a particular total debt ratio or equity multiplier that they view as optimal.
- With this in mind we now consider how rapidly a firm can grow if:
  - it wishes to maintain a particular total debt ratio; and
  - it is unwilling to sell new stock

Given these assumptions, the maximum growth rate that can be achieved is called the Sustainable Growth Rate:

$$\text{Sustainable Growth Rate} = \frac{ROE \times b}{1-(ROE \times b)}$$

$$\text{Sustainable Growth Rate} = \frac{0.14 \times \frac{2}{3}}{1-\left(0.14 \times \frac{2}{3}\right)}$$

Sustainable Growth Rate = 10.29%

The reason for Sustainable Growth rate (10.29%) being larger than internal growth rate (7.23%) is that, as the firm grows, it will have to borrow additional funds if it has to maintain a constant debt ratio. This new borrowing is an extra source of financing in addition to internally generated funds, so A2Z can expand more.

Determinants of Growth

- Using Du Pont identity, we saw that ROE can be decomposed into its various components
- Since ROE appears so prominently in determination of Sustainable growth rate, the factors determining ROE are also important determinants of growth.
- We know that:
- ROE = Profit Margin x Total Assets Turnover x Equity Multiplier
- Anything that increases ROE will increase the Sustainable growth rate. Increasing the retention ratio will have the same effect.
- So putting it all together, the firm's ability to sustain growth depends explicitly on the four factors:
  - Profit Margin
  - Total Assets Turnover
  - Financial Policy
  - Dividend Policy

Profit margin

An increase in profit margin will increase the firm's ability to generate funds internally and thereby increase its sustainable growth.

Total Assets Turnover

An increase in firm's total assets turnover increases the sales grow and thereby increase the sustainable growth rate. Increasing total assets turnover is the same thing as decreasing capital intensity.

Financial Policy

An increase in the debt-equity ratio increases the firm's financial leverage. Since this makes additional debt financing available, it increases the sustainable growth rate.

Dividend Policy

A decrease in the percentage of net income paid out as dividends will increase the retention ratio. This increase internally generated equity and thus increases internal and sustainable growth.

- The sustainable growth rate illustrates the explicit relationship between the firm's four major areas of concern:
  - Operating efficiency (as measured by profit margin)
  - Asset use efficiency (as measured by total assets turnover)
  - Financial policy (as measured by the debt-equity ratio)
  - Dividend policy (as measured by the retention ratio)

If sales are to grow at a rate higher than the sustainable growth rate, the firm must:

- increase profit margins,
- increase total assets turnover,
- increase financial leverage,
- increase earnings retention, or
- Sell new shares

Highlighted by Mir Shahzaz

LESSON 12

USING FINANCIAL STATEMENTS INFORMATION

- Now we take a look at some practical aspects of the financial statements analysis.
  - Reasons for doing financial statements analysis
  - Benchmarking the information
  - Problems arising in the process

Why Evaluate Financial Statements

- Primary reason for looking at the accounting information is that we don't have and can't expect to get market value information. But if we have such information, we will use it instead of accounting data.
- If there is a conflict between accounting and market data, market data would be preferred.

Why Evaluate Financial Statements

- Financial statements analysis is an application of management by exception and boils down to comparing ratios for one business with some average or representative ratios.
- The ratios differing considerably from averages are studied further.

Internal Uses

- Performance Evaluation
  - Profit margin and return on equity
  - Comparing the performance of different divisions
- Planning for the future
  - Historical information used for generating projections
  - Checking the realism of assumptions for the projections

External Uses

- Customers:
  - To evaluate the credit standing of a new customer
  - Large customers would eye on the sustainability of the firm
- Suppliers:
  - Evaluate the financial worth of the supplier
  - Suppliers would be concerned about the creditworthiness of the firm

External Uses

- Competitors
  - Potential strength of the competitors in case of a new product launched by a firm
- Acquisition of new firms
  - Identification of potential targets
  - What to offer.

Choosing a Benchmark

- Benchmarking is to establish a standard to follow for comparison.
- Some methods of benchmarking are:
  - Time-Trend analysis
  - Peer Group Analysis

Highlighted by MR Shahbaz

Time-Trend Analysis

- Based on the historical data of the firm
  - If the current ratio of a firm is 2.4 for the recent financial statements, we may compare it with the current ratios for last 10 years.
  - We may find that current ratio has declined over the years because of
    - More efficient usage of current assets
    - Change in the nature of business of the firm
    - Change in business practices of the firm

Peer Group Analysis

Identifying the firms

- competing in the same markets,
- Having similar assets,
- Operate in similar ways

Benchmarking:

- averages for this group of firms OR
- the top firms among the group

Problems with Financial Statements Analysis

- No underlying theory to help identify the items or ratios to look at or to guide in establishing benchmark
- Very little help on value and risk
  - Which ratios matter the most?
  - What a high or low value might be?
- Firms with many diversified businesses
- Different accounting standards and procedures in different parts of the world

Time Value of Money

- It refers to the fact that a dollar in hand today is worth more than a dollar promised at some time in future.
- The trade-off between money today and money later depends on, among other things, the rate one can earn by investing the money today for some interest income.

Simple Interest vs. Compound Interest

Its most basic form, interest is calculated by multiplying principal (amount invested) by rate (% of interest) multiplied by time (number of periods the interest is calculated). This is called simple interest.

$$I = P \times r \times t$$

Where
P => principal amount
r => interest rate
t => time periods (years)
I => simple interest

However, if interest is left in the account to accumulate for a longer period (usually longer than one year), common practice requires that after interest is earned and credited for a given period, the new sum of principal + interest must now earn interest for the next period, etc. This is compound interest.

$$I = P \times r^t$$

Highlighted by MR Shahbaz

Future Value

- It refers to the amount of money an investment will grow to over some period of time at some given interest rate.
- Alternatively, future vale is the cash value of an investment at some time in future.
- Simple Interest is calculated as:

$$I = P \times r \times t$$

A $1,000 deposit at 8% per year for 3 years' simple interest:
I = 1000 x .08 x 3 = 240
A $1000 deposit at 8% simple interest for three years earns $240 interest.

- The **future value (FV)** of a simple interest calculation is derived by adding the original principal back to the interest earned.

$1,000 + $240 = $1,240

Expressed as a formula:

FV = P(1 + rt)
FV = 1000+(1000 x .08 x 3) = 1,240
In the one-period case, the formula for FV can be written as:
$FV = C_0 \times (1 + r)$
Where C_0 is cash flow today (time zero) and 'r' is the appropriate interest rate.

Future Value for a Lump Sum

If $100 is invested at 10% interest rate, the future value of this $100 in each proceeding year would be:

1. $110 = $100 x (1 + .10)
2. $121 = $110 x (1 + .10) = $100 x 1.10 x 1.10
   = $100 x $(1.10)^2$
3. $133.10 = $121 x (1 + .10) = $100 x 1.10 x 1.10 x .10
   = $100 x $(1.10)^3$

The Multi-period Case: Future Value

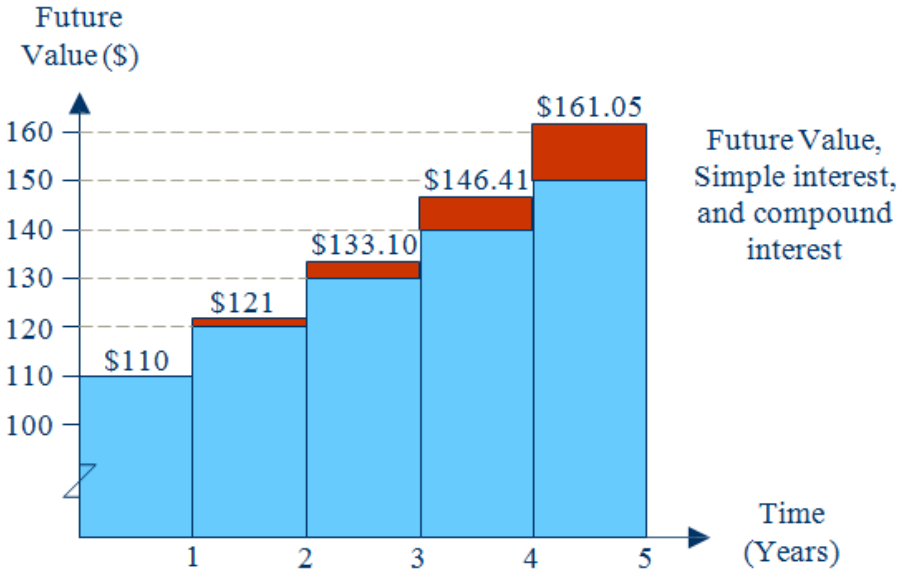
Generalizing the future value of an investment over many periods:
$FV = C_0 \times (1 + r)t$
Where, C_0 is cash flow at date 0,
r is the appropriate interest rate, and
t is the number of periods over which the cash is invested. The expression $(1 + r)^t$ is the *future value interest factor*.

The Multi-period Case: Future Value

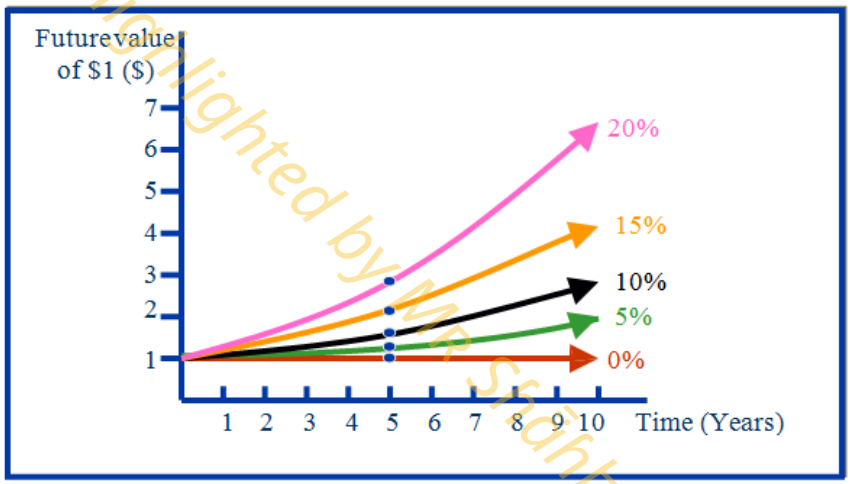
Year	Beginning Amount	Interest Earned	Ending Amount
1	$100.00	$10.00	$110.00
2	110.00	11.00	121.00
3	121.00	12.10	133.10
4	133.10	13.31	146.41
5	146.41	14.64	161.05
	Total Interest	**61.05**	

Highlighted by MP Shahbaz

The Multi-period Case: Future Value

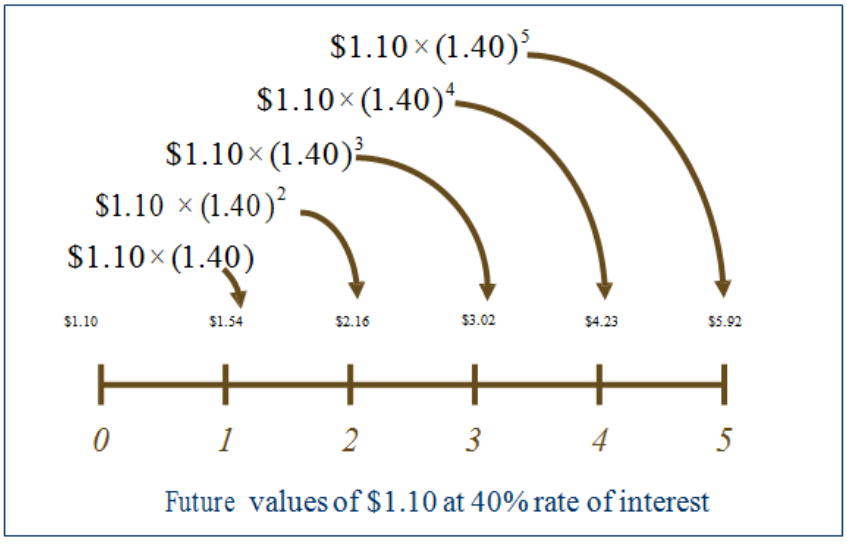


Future Value Projections



Highlighted by MR Shahzad

Future Value and Compounding



Future Value Interest Factors (FVIF)

Number of periods	Interest Rates			
	5%	10%	15%	20%
1	1.0500	1.1000	1.1500	1.2000
2	1.1025	1.2100	1.3225	1.4400
3	1.1576	1.3310	1.5209	1.7280
4	1.2155	1.4641	1.7490	2.0736
5	1.2763	1.6105	2.0114	2.4883

Highlighted by MR Shahbaz

LESSON 13

FUTURE VALUE

- Usually simple interest is used in financial institutions for interest periods of less than one year.
- If the rate is expressed as an annual rate (normal practice), then the time period (t) must be a fraction of a year.
- By investing $10,000 in an 8%, 90-day certificate of deposit, total proceeds at the end of the CD period will be:
- FV = (10,000)+(10,000x.08x90/365) = $10,197.26

If you were to invest $10,000 at 5-percent interest for one year, your investment would grow to $10,500

$500 would be interest ($10,000 × .05)
$10,000 is the principal repayment ($10,000 ×1)
$10,500 is the total due. It can be calculated as:

$10,500 = $10,000 × (1.05)

$10,000 today is worth $10,500 in one year, given that interest rate is 5%

Present Value

Highlighted by MR Shahzad

- It refers to the current value of the future cash flow discounted at the appropriate discount rate.
- In other words, the amount one would need to invest today at some pre-determined interest rate to get some desired amount in future is the present value of the desired money.
- Often, if a bank or other financial institution loans a sum for a short term, the lender will prefer to calculate the interest up front and loan out the *discounted* principal, or principal minus interest to be earned.
- The interest to be paid up front on a loan is called *discount* and the discounted principal, or the actual amount loaned is called the *present value* (PV)

$$PV = \frac{FV}{(1+rt)}$$

- Repeating the discount basic formula (simple interest):

$$PV = \frac{FV}{(1+rt)}$$

- Example: If the bank loans out $10,000 for 90 days at 8% simple interest, the PV is:

PV = 10000 / [1 + (.08)(90/365)]
= 10000/ 1.019726
= $9,806.56

Present Value

Suppose you need $400 to buy textbooks next year and you can earn 7% on your money. How much do you have to put up today? Now, ---------- Present value x 1.07 = $400. Solving for present value:

$$\text{Present Value} = \frac{\$400}{(1.07)} = \$373.83$$

So investing $373.83 (present value) at 7 %will result in having $400 (future value) in one year. If you were to be promised $10,000 due in one year when interest rates are at 5-percent, your investment be worth $9,523.81 in today's dollars.

$$\$9{,}523.81 = \frac{\$10{,}000}{(1.05)}$$

So, the amount that a borrower would need to set aside today to, be able to meet the promised payment of $10,000 in one year is called the Present Value (PV) of $10,000.Note that ------- $10,000 = $9,523.81×(1.05)

In the one-period case, the formula for PV can be written as:

- Where C1 is cash flow at date 1 and
- r is the appropriate interest rate or discount rate.

$$PV = \frac{C_1}{1+r}$$

Present Value for Multiple Periods

- Calculating present value for multiple periods is quite similar in nature as was in case of future value.
- General formula for calculating present value of C cash flow in t periods time is:

$$PV = C \times \frac{1}{(1+r)^t}$$

- $1/(1 + r)^t$ is used to discount a future cash flow, so it is called the ***discount factor*** Or ***present value interest factor (PVIF r,t)***

Calculating the present value of a future cash flow to determine its worth today is commonly called **discounted cash flow (DCF) valuation**

Highlighted by MR Bilal

Present Value Interest Factors (PVIF)

Years	Rate of Interest			
	5%	10%	15%	20%
1	0.9524	0.9091	0.8696	0.8333
2	0.9070	0.8264	0.7561	0.6944
3	0.8638	0.7513	0.6575	0.5787
4	0.8227	0.6830	0.5718	0.4823
5	0.7835	0.6209	0.4972	0.4019

Present Value for Multiple Periods

- Do you want to be a millionaire? No problem!
- Suppose you are currently 21 years old, and can earn 10 percent on your money.
- How much you must invest today in order to accumulate $1 million by the time you reach age 65?

First define the variables:

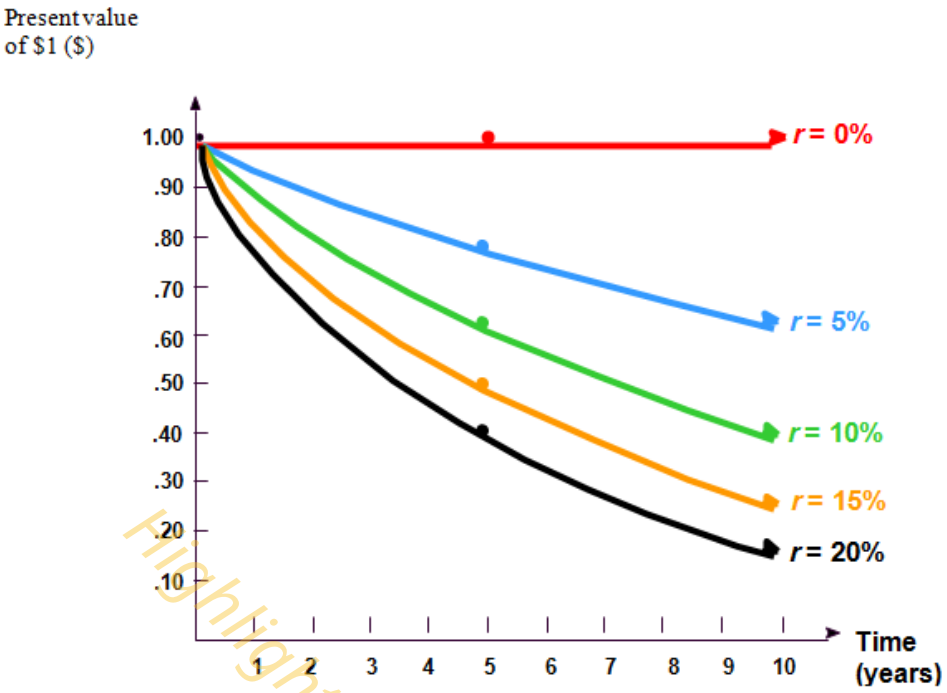
FV = $1 million r = 10 percent
t = 65 - 21 = 44 years PV = ?

Set this up as a future value equation and solve for the present value:

$1 million = PV x (1.10)44
PV = $1 million/(1.10)44 = $15,091

Of course, we've ignored taxes and other complications, but stay tuned - right now you need to figure out where to get $15,000!

Present Value of $1 for Different Periods and Rates



Highlighted by MR Shahzad

Present Value

How much would an investor have to set aside today in order to have $20,000 five years from now if the current rate is 15%?



Present Value vs. Future Value

- What we called the present value factor is just the reciprocal of the future value factor.
  - Future value factor = $(1 + r)^t$
  - Present value factor = $1/(1 + r)^t$

If we let FV_t stand for the future value after t periods, then the relationship between the future value and the present value is:

$$PV \times (1 + r)^t = FVt$$
$$PV = FV_t / (1 + r)^t = FV_t \times [1/ (1 + r)^t]$$

This is also known as basic present value equation.

LESSON 14

EVALUATING INVESTMENTS

- A company is considering the purchase of an asset for $335, which may be sold for $400 in three years. If the discounting factor is 10%, is this a good investment?
- Since the company can invest $335 elsewhere at 10%, in three years, it would grow to:

$$\$335 \times (1 + r)^t = \$335 \times 1.1^3$$
$$= \$335 \times 1.331$$
$$= \$445.89$$

- Since the proposed purchase of asset pays out only $400, it is not as good as other investment alternatives
- Another way of looking at it is to calculate the present value of $400 in three years at 10%:

$$\$400 \times [1/(1+r)^t]$$
$$= \$400 / 1.1^3$$
$$= \$400 / 1.331$$
$$= \$300.53$$

This tells us that we only have to invest about $300 to get $400 in three years, not $335

Highlighted by MR Shahzad

What Rate Is Enough?

- Assume the total cost of a college education will be $50,000 when a child enters college in 12 years.
  The Parents have $5,000 to invest today.
- What rate of interest must you earn on your investment to cover the cost of your child's education?

$$FV = C \times (1+ r)^t$$
$$\$50,000 = \$5,000 \times (1+ r)^{12}$$
$$(1+ r)^{12} = \$50,000 / \$5,000$$
$$(1+ r)^{12} = 10$$
$$(1+ r)^{1/12} = (10)^{1/12}$$
$$1 + r = 1.2115$$
$$r = 1.2115 – 1$$
$$r = .2115$$

Franklin: A Case Study

- Benjamin Franklin died on April 17, 1790. In his will, he gave 1,000 pounds sterling to Massachusetts and the city of Boston. He gave a like amount to Pennsylvania and the city of Philadelphia.
- The money was paid to Franklin when he held political office, but he believed that politicians should not be paid for their service.
- Franklin originally specified that the money should be paid out 100 years after his death and used to train young people.
- Later, however, after some legal wrangling, it was agreed that the money would be paid out 200
- years after Franklin's death in 1990.
- By that time, the Pennsylvania bequest had grown to about $2 million; the Massachusetts bequest had grown to $4.5 million. The money was used to fund the Franklin Institutes in Boston and Philadelphia.
- Assuming that 1,000 pounds sterling was equivalent to 1,000 dollars, what rate did the two states earn? (Note: the dollar didn't become the official U.S. currency until 1792.)

- For Pennsylvania, the future value is $ 2 million and the present value is $ 1,000. There are 200 years involved, so we need to solve for r in the following:

$ 1,000 = $ 2 million /(1 + r)^{200}
$(1 + r)^{200} = 2{,}000.00$

Solving for r, the Pennsylvania money grew at about 3.87% per year. The Massachusetts money did better; check that the rate of return in this case was 4.3%. Small differences can add up!

Finding the Number of Periods

If we deposit $5,000 today in an account paying 10%, how long does it take to grow to $10,000?

$$FV = C_0 \times (1+r)^t$$

$$\$10{,}000 = \$5{,}000 \times (1+0.10)^t$$

$$\frac{\$10{,}000}{\$50{,}00} = (1+0.10)^t$$

$$(1.10)^t = 2$$

$$ln(1.10)^t = ln2$$

$$t = \frac{\ln 2}{ln(1.10)}$$

$$t = \frac{0.6931}{0.0953}$$

$$t = 7.27 \text{ Years}$$

Highlighted by MR Shahbaz

Finding the Number of Periods: Rule of 72

- For reasonable rates of return, the time it takes to double the money, is given approximately by

t = 72 / r%

- Continuing with the example, we have discount rate of 10%, so:

t = 72 / 10 = 7.2 years

- This rule is fairly applicable to discount rates in 5% to 20% range.

Finding the Number of Periods: An Example

- A company has been saving up to purchase an asset. Total cost will be $10 million. The company has currently about $2.3 million.
- How long it will have to wait,
  - if it can earn 5% on the money?,
  - if it can earn 16% on the money?
- At 5% the company will have to wait for a long time:

$2.3 = $10 / 1.05t
1.05t = 4.35
t = 30 years

- At 16 %, it will make it in 10 years.

Summarizing Time Value Calculations

I. Symbols:

PV = Present value, what future cash flows are worth today
FVt = Future value, what cash flows are worth in the future
r = Interest rate, rate of return, or discount rate per period
t = number of periods
C = cash amount

II. Future value of C dollars invested at r percent per period for t periods:

$FV_t = C \times (1 + r)^t$
The term $(1 + r)^t$ is called the *future value factor*.

III. Present value of C to be received in t periods at r percent per period:

$PV = C/(1 + r)^t$
The term $1/(1 + r)^t$ is called the *present value factor*.

IV. The basic present value equation giving the relationship between present and future value is:

$PV = FV^t/(1 + r)^t$

Valuation of Multiple Cash Flows

- So far we have restricted our attention to either future value of a lump-sum present amount or the present value of some single future cash flows.
- Now onwards we will focus on valuation of multiple cash flows.

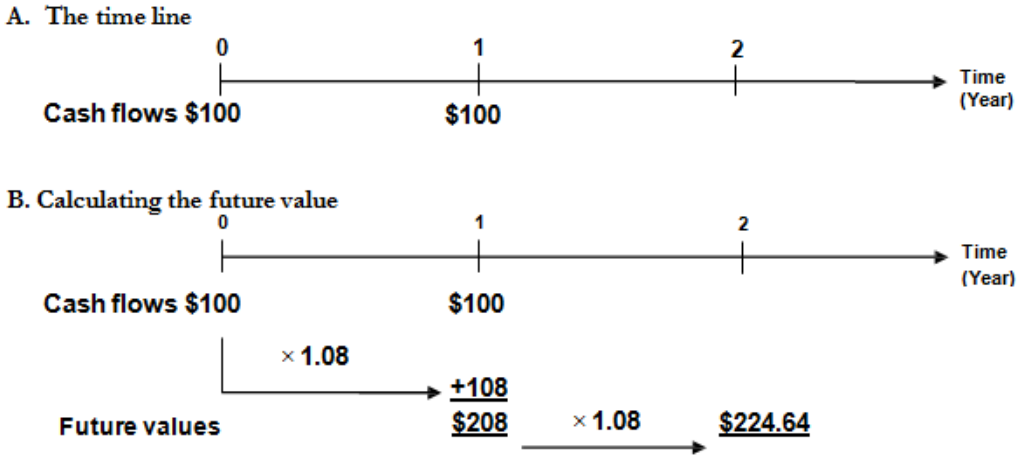
Future Value with Multiple Cash Flows

- If you deposit $100 today in an account at 8% and deposit another $100 in one year, how much will you have in two years?

- At the end of first year, you will have $108 plus the second $100 you deposit, amounting to $208.
  By depositing this amount at 8%, what you will get at the end of 2nd year is:
  $208 x 1.08 = $224.64

- This can be better explained by using a time line.

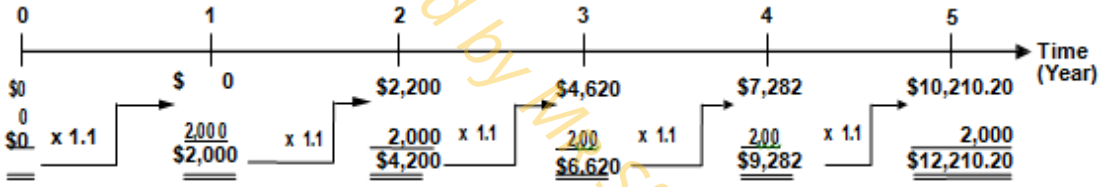
Highlighted by MR Shahbaz



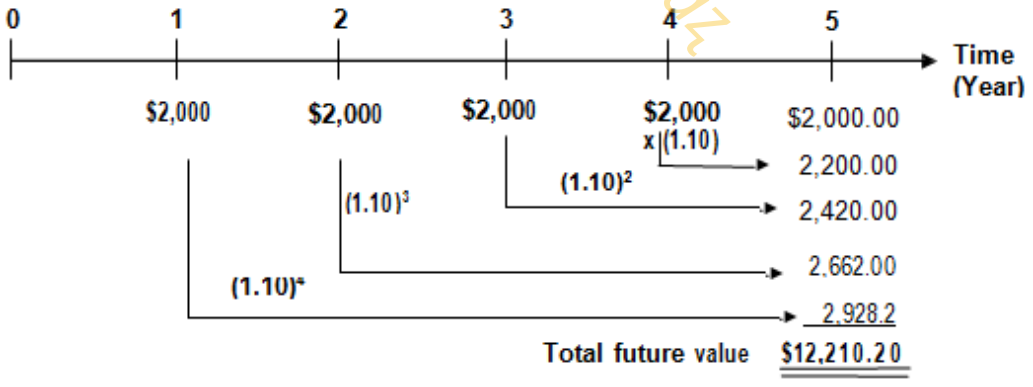
An alternate way to calculate the future value is to calculate the future value of each cash flow separately.

- The future value of first $100 in 2 years at 8% is:
  FV = $100 x 1.08^2 = $100 x 1.1664 = $116.64
- The 2nd $100 is deposited for one year, thus:
  FV = $100 x 1.08 = $108
- Adding up both the future values we get the total future value:
  $116.64 + $108 = $224.64

Future value calculated by compounding forward one period at a time



Future value calculated by compounding each cash flow separately

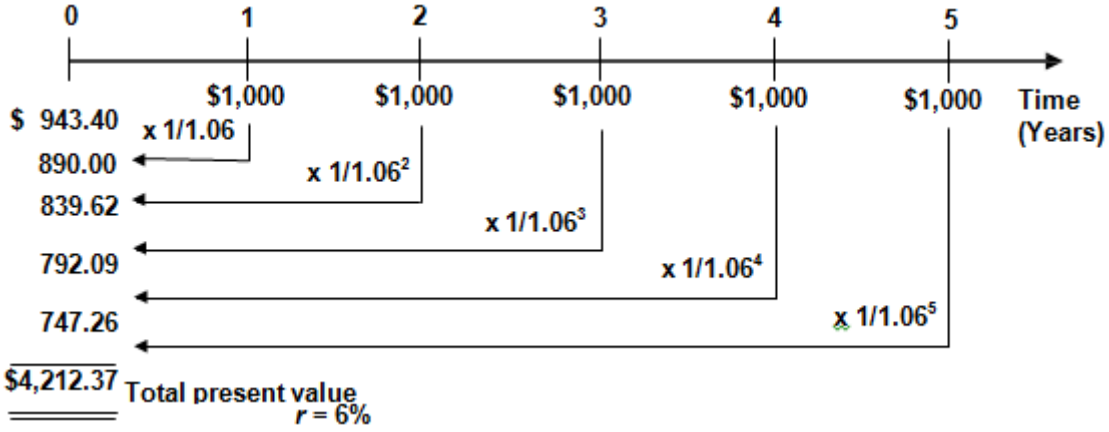


Present Value with Multiple Cash Flows

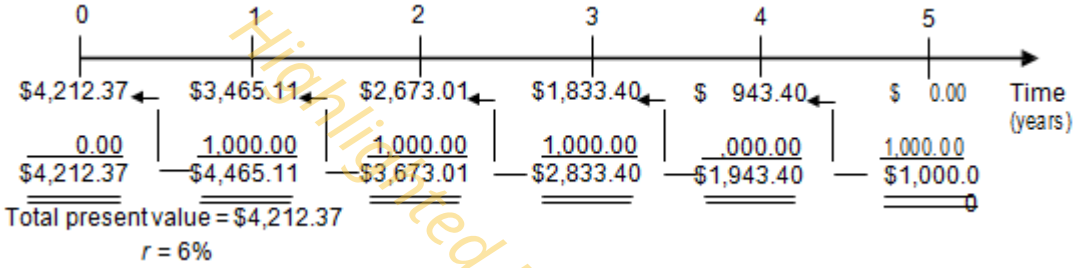
- Suppose, we had an investment that was going to pay $1,000 at the end of every year for next five years. The discount rate is assumed as 6%.
- Present value of all these cash inflows can be calculated by:
- Discounting each cash flows separately OR
- Discounting back one period at a time.

Highlighted by Mashoodiqbal

Present value calculated by discounting each cash flow separately



Present value calculated by discounting back one period at a time



How much is it worth?

- You are offered an investment that will pay you $200 in one year, $400 the next year, $600 in 3rd and $800 at the end of 4th year.
- If you can earn 12% on very similar investments, what is the most you should pay for this one?
- Calculating the present value of each cash flow at 12%:

$200 x 1/1.12¹ = $200 / 1.1200 = $178.57
$400 x 1/1.12² = $400 / 1.2544 = 318.88
$600 x 1/1.12³ = $600 / 1.4049 = 427.07
$800 x 1/1.12⁴ = $800 / 1.5735 = 508.41
Total Present Value=$1,432.93

- If you can earn 12% on your money then you can duplicate this investment's cash flows for $1,432.93. So, this is the most you should be willing to pay.

Annuities and Perpetuities

- We will frequently encounter situations where we have multiple cash flows that are all the same amount.
- Series of equal installments for a loan-repayment
- A series of constant, or level, cash flows that occur at the end of each period for some fixed number of periods is called an ordinary **Annuity**.

Highlighted by Mr Shahzaiz

Present Value for Annuity cash flows

- For annuity calculation, we use a variation of present value equation.
- The present value of an annuity of *C* dollars per period for t periods when interest rate is r is:

$$PV = C \times \frac{\{1 - \text{Present value interest factor}\}}{r}$$

$$PV = C \times \frac{\left\{1 - \frac{1}{(1+r)^t}\right\}}{r}$$

Where,
C = Periodic payment or annuity
r = rate of interest
t = number of periods
The term in the parenthesis is called present value interest factor of an annuity (PVIFAr,t).

Highlighted by MR Shahbaz

LESSON 15

ANNUITIES

- We will frequently encounter situations where we have multiple cash flows that are all the same amount.
- Series of equal installments for a loan-repayment
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$$PV = C \times \frac{\left\{1 - \frac{1}{(1+r)^t}\right\}}{r}$$

Where, C = Periodic payment or annuity
r = rate of interest
t = number of periods
The term in the parenthesis is called *present value interest factor of an annuity*.

Highlighted by Mashhad

Present Value Interest Factor of an Annuity (PVIFA)

Years	Rate of Interest			
	5%	10%	15%	20%
1	0.9524	0.9091	0.8696	0.8333
2	1.8594	1.7255	1.6257	1.5278
3	2.7232	2.4869	2.2832	2.1065`
4	3.5460	3.1699	2.8550	2.5887
5	4.3259	3.7908	3.3522	2.9906

How much can you afford?

- By looking at your budget you know you can pay $632 for a new car, bank is offering you a loan for 48 months at 1% per month. How much should you borrow?

$$PVA = C \times \frac{\{1 - \text{Present value interest factor of Annuity}\}}{r}$$

$$PVA = \$632 \times \frac{\{1 - 0.6203\}}{0.01}$$

$$PVA = \$632 \times 37.9740$$

$$PVA = 24{,}000$$

- So, Present value = $632 x 37.9740 = $ 24,000
- Therefore, you can afford to borrow $24,000

Finding the Payment

- If you want to buy a new car costing $23,000. With a 10% down payment, the bank will loan you the rest at 9% per year (.75% per month) for 60 months. How much will each monthly payment be?
- You will borrow 0.90 x $23,000 = $20,700. This is the amount today, so it's the PV. The rate is 0.09/12 = 0.0075, and there are 60 periods:

$$PVA = C \times \frac{\left\{1-\frac{1}{(1+r)^t}\right\}}{r}$$

$$\$20{,}700 = C \times \frac{\left\{1-\frac{1}{(1+0.0075)^{60}}\right\}}{0.0075}$$

$$\$20{,}700 = C \times 48.1737$$

$$C = \frac{\$20{,}700}{48.1737}$$

C = $429.70 per month

Finding the number of payments

- To repay a loan of $1,000, Mr. X can only afford to pay $20 per month. Interest rate is 1.5% per month.
- How long will it take to repay the loan?
- Here, PV = $1,000, C = $20 , r = 1.5% per month

$1000 = $20 x (1 – PVF)/0.015
($1,000/20) x 0.015 = 1 – PVF
PVF = 0.25 = 1/ (1 + r)t
1.015t = 1/0.25 = 4

- So how long will it take to quadruple the money?

$1.015^x = 4$ → 93 month or 7.75 years
Calculation details
$1.015^x = 4$
Taking log on both sides
$\log 1.015^x = \log 4$
$x \log 1.015 = \log 4$
$x = \log 4 / \log 1.015$
$x = 0.60206/0.00647$
$x = 93$ months or 7.75 years

Finding the rate

- An insurance company offers to pay you $1,000 per year if you pay $6,710 up front. What rate is applicable in this 10-year annuity?
- Here,

C = $1,000, PV = $6,710 and t = 10, r = ?

$6,710 = $1,000 x (1 – PVF) / r
6.71 = {1- [1 /(1 + r)10]} / r

Highlighted by MR Shahzad

Looking at the PVIFA table for 10 periods, 6.7101 is the value for 8%. So insurance company is offering 8%

Annuity Future Value

$$FVA = C \times \frac{\{\text{Future value interest factor of Annuity} - 1\}}{r}$$

$$FVA = C \times \frac{\{(1+r)^t - 1\}}{r}$$

Future Value for Annuities

- Previously we determined that a 21-year old could accumulate $1 million by age 65 by investing $15,091 today and letting it earn interest (at 10% compounded annually) for 44 years.
- Now, rather than plunking down $15,091 in one chunk, suppose she would rather invest smaller amounts annually to accumulate the million.
- If the first deposit is made in one year, and deposits will continue through age 65, how large must they be?
- Set this up as a FV problem:

$1,000,000 = C x [(1.10)^{44} - 1] / 0.10
C = $1,000,000/652.6408 = $1,532.24
Becoming a millionaire just got easier!

- Unfortunately, most people don't start saving for retirement that early in life. (Many don't start at all!)
- Suppose a 40-year old person has decided it's time to get serious about saving. Assuming that he wishes to accumulate $1 million by age 65, he can earn 10% compounded annually, and will begin making equal annual deposits in one year, how much must each deposit be?
- Set this up as a FV problem:

t = 65 - 40 = 25
FV = $1,000,000

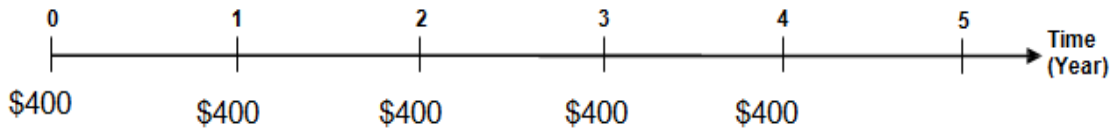
Then:

$1,000,000 = C x [(1.10)$_{25}$ - 1] / 0.10
C = $1,000,000/98.3471 = $10,168.07

- Moral of the story: Putting off saving for retirement makes it a lot more difficult!

Annuities Due

- So far, we have discussed only ordinary annuities, where cash flows occur at the end of each period, e.g. loan repayments
- However, when you lease an asset, the first lease payment is usually due immediately, second at the beginning of second period and so on.
- An Annuity due is an annuity for which cash flows occur at the beginning of each period.
- The time line for Annuity due, having 5 payments of $400 each, would be like:



Highlighted by MR Shahbaz

- Present value of a four year $400 ordinary annuity at 10% is $1,267.95
- Adding on the extra $400, we get $1,667.95, the present value of this annuity due.
- The relationship between an annuity due and an ordinary annuity is just:

Annuity due value = Ordinary annuity value x (1 + r)

Highlighted by MR Shahbaz

LESSON 16

PREPETUITIES

- The present value of a perpetuity is
- Perpetuity PV = C / r
- Suppose we expect to receive $1000 at the end of each of the next 5 years. Our opportunity rate is 6%. What is the value today of this set of cash flows?

$$PV = \$1000 \times (1 - [1/1.06^5]) / 0.06$$
$$PV = \$1000 \times 4.212364$$
$$PV = \$4212.364$$

- Now suppose the cash flow will be $1000 per year forever, making it perpetuity. In this case, the PV is easy to calculate:

$$PV = C/r = \$1000/.06 = \$16,666.67$$

Summary of Annuity and Perpetuity

I. Symbols

PV = Present value, what future cash flows bring today
FV_t = Future value, what cash flows are worth in the future
r = Interest rate, rate of return, or discount rate per period
t = Number of time periods
C = Cash amount

II. FV of C per period for t periods at r percent per period:

$$FVA = C \times \frac{\{(1+r)^t - 1\}}{r}$$

III. PV of C per period for t periods at r percent per period:

$$PVA = C \times \frac{\left\{1 - \frac{1}{(1+r)^t}\right\}}{r}$$

IV. PV of perpetuity of C per period:

$$PV = C / r$$

Effective Annual Rates

- If a rate is quoted as 10% compounded semiannually, then what this means is that the investment actually pays 5% every six months.
- Is 5% every six months the same thing as 10% per year?
- $1 x 1.10 = $1.10
- $1 x 1.05^2 = $1.1025
- 10% compounded semiannually is equivalent to 10.25% compounded annually.
- 10.25% is called effective annual rate (EAR)
- Suppose three banks offer you the following rates for a savings account:
  - Bank A: 15%, compounded daily
  - Bank B: 15.5%, compounded quarterly

Highlighted by MR Shahzad

- Bank C: 16%, compounded annually
- Which one would you opt for:
- Bank C is offering 16% per year, since there is no compounding during the year, this is effective rate.
- Bank B actually paying 0.155/4 = 0.03875 or 3.875% per quarter. An investment of $1 at this rate for 1 year would grow to:

$1 x 1.03875^4 = $1.1642
So, EAR is 16.42%

- For saving, it is good. For borrowing it is worse
- Bank A is compounding daily, so the daily interest rate is actually 0.15 / 365 = 0.000411 or 0.0411%. An Investment of $1 at this rate for 365 days would grow to:

$1 x 1.000411^{365} = $1.1618
So, EAR is 16.18%

- This is not as good as Bank B's 16.42% for a saver, and not as good as Bank C's 16% for a borrower
- Two Lessons:
  - The highest quoted rate is not necessarily the best rate
  - Compounding during the year can lead to a significant difference between the quoted rate and the effective rate
- EAR is computed in three steps
  - Divide the quoted rate by the number of times the interest is compounded
  - Add 1 and raise it to the power of number of times the interest is compounded.
  - Subtract 1

So,

EAR = $(1 + \text{Quoted rate} / m)^m - 1$
Where m is the number of times the interest is compounded

The EAR of 12% compounded monthly is:

EAR = $(1 + \text{Quoted rate} / m)^m - 1$
EAR = $(1 + 0.12/12)^{12} - 1$
EAR = $(1.01)^{12} - 1$
EAR = 1.12685 – 1
EAR = 12.6825%

Highlighted by Mir Shahzad

Effective Annual Rate with changed number of compounding

Compounding period	Number of times compounded	Effective Annual Rate
Year	1	10.000 %
Quarter	4	10.38129
Month	12	10.47131
Week	52	10.50648
Daily	365	10.51558
Hour	8,760	10.51703
Minute	525,600	10.51709

- A bank is offering 12% compounded quarterly. If you put $100 in an account

- How much you will earn at the end of one year?
- What is the EAR?
- How much will you have at the end of two years?
- The bank is offering 12%/4 = 3% every quarter; investing $100 for 4 periods at 3% per period, the future value is:

$$FV = \$100 \times 1.03^4 = \$112.55$$
The EAR is 12.55%

- We can calculate the yield of two years in two ways
- Recognize two years as eight quarters:

$$\$100 \times 1.03^8 = \$126.68$$

- Using EAR for two years:

$$\$100 \times 1.1255^2 = \$126.68$$

- Anytime we do a present value or future value calculation, the rate we use must be an actual or effective rate.

Annual Percentage Rates

- Annual percentage rate (APR) is the interest rate charged per period multiplied by the number of periods per year.
- Laws in some countries require that the lenders disclose an APR on virtually all the consumer loans in a prominent and un-ambiguous way.
- A typical credit card agreement quotes an interest rate of 18% APR. Monthly payments are required. What is the actual interest rate you pay on such a credit card ?
- APR of 18% with monthly payments is really 0.18 / 12 = 0.015 or 1.5% per month.
- So,

$$EAR = (1 + 0.18/12)^{12} - 1$$
$$EAR = 10.15^{12} - 1 = 19.56\%$$

Loans

- Whenever a lender extends a loan, some provision will be made for repayment of the principal amount.
- Here we discuss three forms of repayment of principle and interest patterns.
  - Pure Discount Loans
  - Interest-only Loans
  - Amortized Loans

Pure Discount Loans

- The borrower receives money today and repays a single lump sum at some time in the future.
- A borrower is able to repay $25,000 in 5 years. Given a discount rate of 12%, what amount of money the lender should lend?

$$\text{Present value} = \$25{,}000 / 1.12^5$$
Present Value = $14,186

Highlighted by MR Shahzad

Interest-Only Loans

- Calls for the borrower to pay interest each period and to repay the entire principal at some time in the future.
- With a 3-year, 10%, interest-only loan of $1,000, the borrower would pay $1,000 x 0.10 = $100 in interest at the end of first and second years,
- At the end of third year, he would return $1000 plus the $100 in interest for that year.

Amortized Loans

- Amortized loan requires the borrower to repay parts of the loan amount over time.
- The process of paying off a loan by making regular principal deductions is called amortizing the loan. A loan can be amortized in two ways:
    - Borrower pays the interest each period plus some fixed amount, e.g. medium-term business loans.
    - Borrower makes a single fixed payment every period, car loans and mortgages.

Amortization Schedule - Fixed Principal

A $5000 loan at an interest rate of 9% for 5 years

Highlighted by MR Shahbaz

Year	Beginning Balance	Total Payment	Interest Paid	Principal Paid	Ending Balance
1	$ 5,000	$ 1,450	$ 450	$ 1,000	$ 4,000
2	4,000	1,360	360	1,000	3,000
3	3,000	1,270	270	1,000	2,000
4	2,000	1,180	180	1,000	1,000
5	1,000	1,090	90	1,000	0
	Total	$ 6,350	$ 1,350	$ 5,000	

Amortization Schedule - Fixed Payments

- Again, taking a $5,000 loan at an interest rate of 9% for 5 years; we first need to determine the payment.
- Since this loan's cash flows are in the form of an ordinary annuity. So, solving for payments:

$$\$5{,}000 = C \times (1 - 1/0.09^5) / 0.09$$
$$\$5{,}000 = C \times (1 - 0.6499) / 0.09$$
$$C = \$5{,}000 / 3.8897$$
$$C = \$1{,}285.46$$

Amortization Schedule - Fixed Payments

Beginning Year	Total Balance	Total Payment	Interest Paid	Principal Paid	Ending Balance
1	$ 5,000	$ 1,285.46	$ 450	$ 835.46	$ 4,164.54
2	4,164.54	1,285.46	374.81	910.65	3,253.89
3	3,253.88	1,285.46	292.85	992.61	2,261.28
4	2,261.27	1,285.46	203.51	1,081.95	1,179.32
5	1,179.32	1,285.46	106.13	1,179.23	0
	Total	$ 6,427.30	$ 1,427.30	$ 5000	

LESSON 17

BONDS

- An evidence of debt issued by a corporation or a governmental body.
- When a corporation (or government) wishes to borrow from public on a long term basis, it does so by issuing or selling debt securities generally called bonds
- A bond represents a loan made by investors to the issuer. In return for his/her money, the investor receives a legal claim on future cash flows of the borrower.
- The issuer promises to:
  - Make regular coupon payments every period until the bond matures, and
  - Pay the face/par/maturity value of the bond when it matures.
- Default - since the abovementioned promises are contractual obligations, an issuer who fails to keep them is subject to legal action on behalf of the lenders (bondholders).
- B Corporation:
  - Wants to borrow $1,000 for 30 years at 12 % interest rate
  - Will pay 0.12 x $1,000 = $120 in interest every year for 30 years.
  - Will repay $1,000 at the end of 30 years
- B Corporation
  - $120 regular interest payments are the bond's coupons
  - $1,000 is the par value or face value of the bond
  - Annual coupon divided by the par value ($120/$1000 = 12%) is the coupon rate
  - 30 years is the maturity time

Highlighted by MPShahbaz

Bond Values and Yields

- The value of bonds may fluctuate as the interest rates change by time in the market place, though the cash flows from a bond remain the same.
- When interest rates rise, the present value of the bond's remaining cash flows decline and the bond is worth less
- When the interest rates fall, the bond is worth more.
- To determine the value of bond at a particular point in time, we need to know:
  - No. of periods remaining till maturity,
  - The face value,
  - The coupon rate, and
  - The market interest rate for similar bonds
- The interest rate required in the market on bonds is called the bond's Yield to Maturity
- The X Corporation issues a bond with 10 years to maturity having annual coupon of $80. Similar bonds have a yield to maturity of 8%.
- X bond's cash flows have two components:
  - **an annuity component (coupons) and**
  - **a lump sum (face value paid at maturity)**
- The X Corporation
- At the going interest rate of 8% the present value of $1,000 paid in 10 years is:

$$PV = \$1,000 / 1.08^{10} = \$1,000 / 2.1589 = \$463.19$$

- Present value of the annuity of 80$ per year for 10 years is:

$$PV = \$80 \times (1 - 1/1.08^{10}) / 0.08$$
$$PV = \$80 \times 6.7101$$
$$PV = \$536.81$$

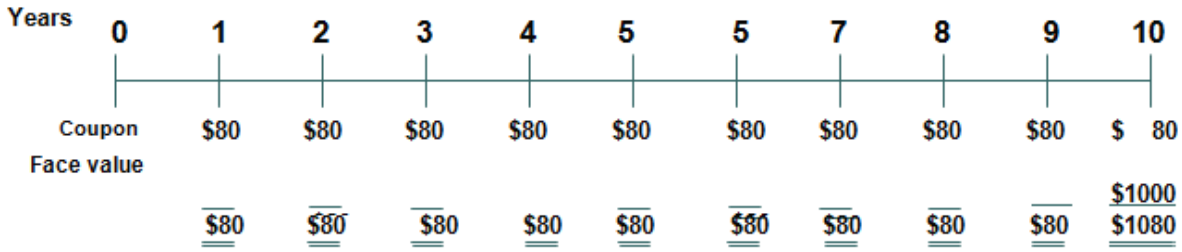
The X Corporation

- To get the bonds value we add up both parts

Total Bond Value = $463.19 + $536.81
Total Bond Value = $1,000

- This means that the bond sells for exactly its face value.

Alternatively,



- Interest rate change
- Interest rate risen to 10% after one years (9 years to maturity)
- **Now the present value of $1,000 paid in nine years at 10% is**

$$\$1{,}000 / 1.10^9 = \$1{,}000/2.3579 = \$424.10$$

- **And present value of $80 annuity for 9 years at 10% is**

$$\$80 \times (1 - 1/1.10^9)/0.10 = \$80 \times 5.7590 = \$460.72$$

- Adding both parts:

Total bond value is $424.10 + 460.72 = $884.82

- Therefore, the bond should sell for about $885
- Because the bond sells for less than the going rate, investors are willing to lend something less than $1,000.
- Because the bond sells for less than face value, it is said to be a discount bond.
- The investor who purchased and kept bond would get $80 per year and would have a $115 gain at maturity as well. This gain compensates the lender for below-market coupon rate.

Another way to see why bond is discounted by $115 is to note that the $80 coupon is $20 below the coupon on a newly issued par value bond. So the investor who buys and keeps the bond gives up $20 every year for 9 years. At 10 % this annuity is worth:

$$\$20 \times (1 - 1/1.10^9)/0.10 = \$20 \times 5.7590 = \$115.18$$

Just as rise of interest rates reflected a decline in the price of the bond, a drop of 2% in interest rates would result in the bond being sold for more than $1000. Such a bond is said to sell at a premium or is called a premium bond.

Highlighted by MP Shahzad

LESSON 18

VALUING A BOND

- Assume you have the following information.
  - BMN, Inc. bonds have a $1000 face value
  - The promised annual coupon is $100
  - The bonds mature in 20 years
  - The market's required return on similar bonds is 10%

- Present value of the face value

$$= \$1000 \times [1/1.10^{20}]$$
$$= \$1000 \times 0.14864$$
$$= \$148.64$$

- Present value of the coupon payments

$$= \$100 \times [1 - (1/1.10^{20})]/0.10$$
$$= \$100 \times 8.5136$$
$$= \$851.36$$

- The value of each bond = $148.64 + 851.36
  = $1000

Valuing a Bond: A Discount Bond

- Assume you have the following information.
  - BMN, Inc. bonds have a $1000 face value
  - The promised annual coupon is $100
  - The bonds mature in 20 years
  - The market's required return on similar bonds is 12%

- Present value of the face value

$$= \$1000 \times [1/1.12^{20}]$$
$$= \$1000 \times 0.10366$$
$$= \$103.66$$

- Present value of the coupon payments

$$= \$100 \times [1 - (1/1.12^{20})]/0.12$$
$$= \$100 \times 7.4694$$
$$= \$746.94$$

- The value of each bond = $103.66 + 746.94
  = $850.60

Valuing a Bond: A Premium Bond

- Assume you have the following information.
  - BMN, Inc. bonds have a $1000 face value
  - The promised annual coupon is $100
  - The bonds mature in 20 years
  - The market's required return on similar bonds is 8%

Highlighted by MR Shahbaz

- Present value of the face value

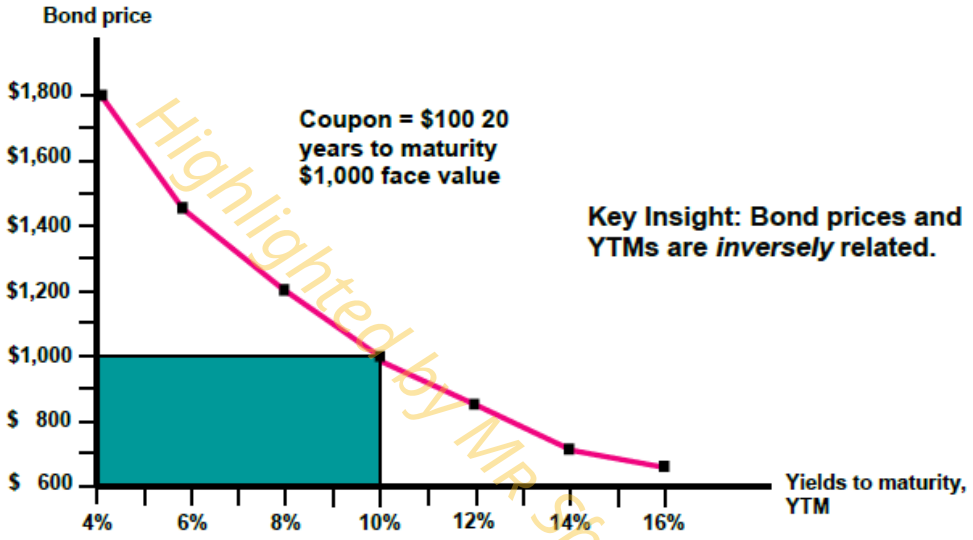
$$= \$1000 \times [1/1.08^{20}]$$
$$= \$1000 \times 0.21455$$
$$= \$214.55$$

- Present value of the coupon payments

$$= \$100 \times [1 - (1/1.08^{20})]/0.08$$
$$= \$100 \times 9.8181$$
$$= \$981.81$$

- The value of each bond = \$214.55 + 981.81
  = \$1,196.36

Bond Price Sensitivity to YTM



Valuing Bonds

Based on our examples we can write a general expression for the value of a bond. If a bond has

- a face value of F paid at maturity
- a coupon of C paid per period
- t periods to maturity
- a yield of r per period

Its value is:

$$\text{Bond Value} = C \times \frac{\left\{1-\frac{1}{(1+r)^t}\right\}}{r} + \frac{F}{(1+r)^t}$$

Where,

- $C \times \frac{\left\{1-\frac{1}{(1+r)^t}\right\}}{r}$ is the present value of coupons, and
- $\frac{F}{(1+r)^t}$ is the present value of face amount?

Highlighted by MP Shahzad

Semiannual Coupons

- Bond yields are quoted like APRs, the quoted rate is equal to the actual rate per period multiplied by the number of periods in a year.
- When the payments of coupons are made on semiannually, each payment of half of the annual coupon, it is referred to as semiannual coupon bond
- If an ordinary bond has a coupon rate of 14% then the owner gets $140 per year but in two installments each of $70. in this case, the Yield to maturity is 16%
- So with a 16% quoted yield, the true yield is 8% per period.
- Now if the bond has seven years to maturity
- What would be the bond's price?
- What is the effective annual yield on this bond?
- The bond should sell at a discount because it has a coupon rate of 7% per six months when the market requires 8% every six months.

- Present value of the face value

$$= \$1000 / 1.08^{14}$$
$$= \$1000/2.9372$$
$$= \$340.46$$

- Present value of 14 period annuity of $70 at 8% discount rate is

$$= \$70 \times [1 - (1/1.08^{14})] /0.08$$
$$= \$70 \times 8.2442$$
$$= \$577.10$$

- Total Present value tells that the bond should sell for:

$$= \$340.46 + 577.10$$
$$= \$917.56$$

- To calculate the effective annual yield on this bond, note that 8% every six months is equivalent to:

$$EAR = (1 + 0.08)^2 - 1$$
$$EAR = 16.64\%$$

Highlighted by MR Shahbaz

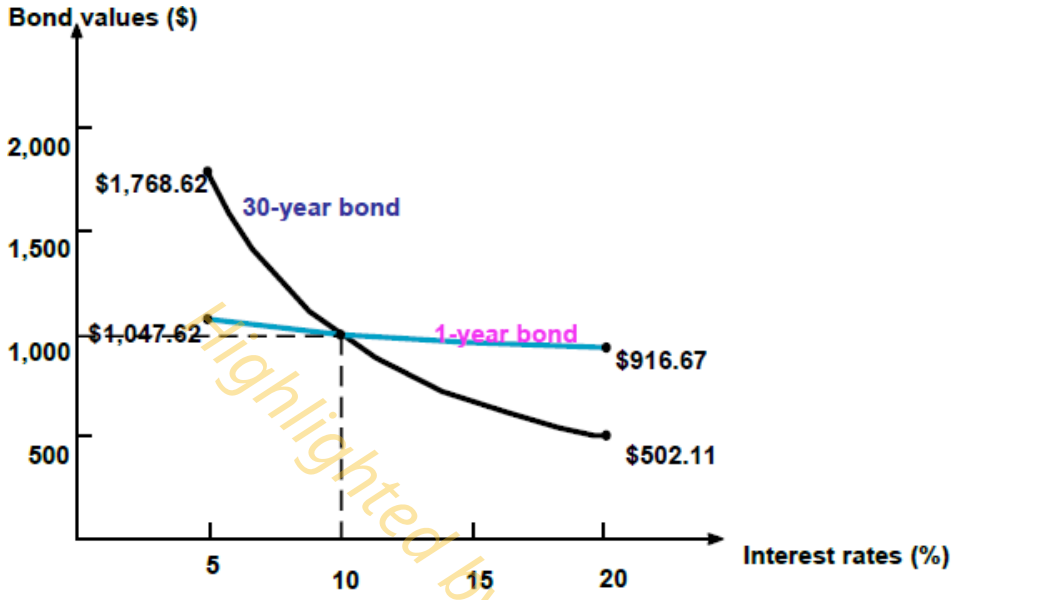
Interest Rate Risk

- The risk that arises for bond owners from fluctuating interest rates is called interest rate risk.
- The risk on a bond depends on how sensitive its price is to interest rate changes
- The sensitivity directly depends on
- Time to maturity
- Coupon rate
- While looking at a bond following should be kept in mind
- All other things being equal, the longer time to maturity, the grater the interest rate risk
- All other things being equal, the lower the coupon rate, the greater the interest rate risk
- We can illustrate these characteristics graphically

Interest Rate Risk

Interest Rate	Time to Maturity	
	1 Year	30 Years
5%	$1047.62	$1768.62
10	1000.00	1000.00
15	956.52	671.70
20	916.67	502.11

Interest Rate Risk



Value of a Bond with a 10% Coupon Rate for Different Interest Rates and Maturities

Highlighted by MR Shahzad

- The slope of the line connecting the prices is much steeper for 30-year maturity than it is for 1-year maturity
- The steepness tells that a relatively small change in interest rate will lead to a substantial change in the bond's value.
- In comparison, one year bond is relatively less sensitive to interest rate changes
- The reason for longer term bonds having greater interest rate sensitivity is that a large proportion of the bond's value comes from the $1000 face amount.
- The present value of this amount is not affected by small change in interest rate if amount is to be received in an year
- While, even a small change in interest rate when compounded for 30 years, will have a significant effect on the present value.
- Interest rate risk increases at a decreasing rate.
- If we compared a 10-year bond with a 1-year bond, obviously, 10 year bond has much greater risk
- But if we compare a 20 year bond to a 30 year bond, the 30 year bond will have a somewhat greater interest rate risk but the difference is fairly small.
- The bonds with lower coupons have greater interest rate risk
- If two bonds with different coupon rates have same maturity, the value of the one with the lower coupon is proportionately more dependent on the face amount to be received. So its value will fluctuate more as interest rate changes
- In other words, bond with higher coupon has a larger cash flow early in its life, so its value is less sensitive to changes in discount rate